Testis, Penis & Urethra

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 78	Anatomy of the Retroperitoneum
Ch 79	Neoplasms of the Testis
Ch 80	Surgery of Testicular Tumours
Ch 81	Minimally Invasive RPLND
Ch 82	Tumours of the Penis
Ch 84	Inguinal Node Dissection
Ch 83	Tumours of the Urethra
Ch 85	Benign Disorders of the Penis & Urethra
Ch 86	Urethral Stricture Disease
Ch 87	Surgery of the Scrotum & Seminal Vesicles
★	Putting it together

78 Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

1 · Definition, boundaries and organs

Why this matters: it shows what lies in your path to the kidney.

- P1 Describe in brief the anatomy of the retroperitoneum… retroperitoneal laparoscopic nephrectomy — *boundaries, primary vs secondary organs, flank layers, lumbodorsal fascia at 12th rib tip, three spaces, lateroconal fascia, perirenal cone open below, fat colour*

Not in the paper answer:

- The **anal canal** is primarily retroperitoneal.
- The medial Gerota–transversalis fusion must be incised to reach the hilum (section 3).
- Whether psoas and quadratus lumborum are boundaries or contents is debated. Secondary organs exclude the first part of the duodenum (**D1**) and pancreatic tail.
- Primary organs lie in the perirenal space; secondary organs in the anterior pararenal space.

> **KEY POINT**
> Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney.

2 · Posterior abdominal wall and surface landmarks

Why this matters: these guide flank incisions and percutaneous access.

- The upper pole is at the 11th rib (left) and 12th (right, lower). The hilum is at **L1**; the lower pole at L3–L4.
- External oblique runs inferomedially ("hands in pockets"); internal oblique runs superomedially.
- **Psoas major** (T12–L5 to lesser trochanter) flexes the hip with iliacus. **Psoas minor** is inconstant and inserts at the pelvic brim. **Quadratus lumborum** lies posterolateral to psoas and stabilises the 12th rib.
- Lower rib fracture suggests retroperitoneal injury. The **costovertebral (lumbocostal) ligament** tethers the 12th rib to the L1–L2 transverse processes; dividing it frees the rib in posterior approaches.
- The **intercostal bundle** lies under the rib above (vein, artery, nerve top-down), so the nerve is least protected. Supracostal access may cross the pleura, so it has more complications than subcostal.
- **Dorsal lumbotomy** incises the lumbodorsal fascia vertically, lateral to erector spinae and quadratus: **no muscle is cut**.

3 · Retroperitoneal fasciae and spaces

Why this matters: collections spread along these.

Three fetal strata form three fasciae and three spaces: outer (transversalis; posterior pararenal), intermediate (Gerota; perirenal, genitourinary) and inner (fusion fascia with dorsal peritoneum; anterior pararenal, gastrointestinal).

- In the **posterior pararenal** space, the posterior Gerota lamina fuses medially with transversalis; this **must be incised to reach the hilum**.

- The Toldt (anterior) lamina is thinner; the Zuckerkandl (posterior) lamina is thicker and seen more on imaging. MCQ Gerota described only the anterior, yet his name covers both.
- The perirenal cone may cross the midline below the hilum. Urinoma, haematoma (ruptured aneurysm) and abscess are contained laterally but **track caudally**. Fat colour change guides the plane in colon mobilisation.
- The **anterior pararenal** space holds colon, pancreas and D2–D3, and is entered from the **white line of Toldt**.

4 · Gastrointestinal relations (operative hazards)

Why this matters: bowel, pancreatic and splenic injuries arise here.

- **Right**: the pancreatic head lies in front of the inferior vena cava (IVC), and D2 lies on the right hilum, hence the **Kocher manoeuvre**. D3 crosses between the superior mesenteric artery (SMA) in front and the aorta.
- **Left**: pancreatic tail, spleen and stomach lie near the upper pole. Splenic vessels on the upper pancreatic border are at risk when the pancreas is lifted off renal fascia.
- Divide hepatocolic/splenocolic ligaments sharply; avoid over-retraction. Close small mesenteric defects (internal hernia).

5 · Great vessels

Why this matters: major operative bleeding comes from these vessels.

Aorta and branches

The great vessel compartment runs from T12 to L4/5. The aorta enters at **T12**, left of the IVC, and bifurcates at **L4**.

Branch	Type	Level	Supplies
Inferior phrenic	Paired lateral	First, below hiatus	Diaphragm; superior adrenal
Celiac	Anterior	T12	Foregut
Middle adrenal	Paired lateral	SMA level	Adrenal
SMA	Anterior	L1–L2	Midgut
Renal	Paired lateral	L1–L2, below SMA	Kidney; inferior adrenal
Gonadal	Paired **anterolateral**	Below renals	Testis/ovary
Lumbar	Paired posterior	Four pairs, L1–L4	Wall, spine, cord
Inferior mesenteric (IMA)	Anterior	L3–L4	Hindgut

- The superior adrenal artery is constant; middle and inferior vary. Supernumerary renal arteries occur in ≥ ¼ *(see Ch 84)*.
- The **testicular artery** runs in front of psoas, genitofemoral nerve, ureter (and right IVC) to the deep ring.
- **Ureteric supply**: proximal renal; middle aorta, common iliac, gonadal; distal internal iliac. The abdominal ureter is supplied **medially**, the pelvic ureter **laterally**.
- IMA ligation is safe: the marginal artery of **Drummond** feeds the colon; superior rectal joins middle/inferior rectal. **SMA ligation is not safe**: it causes severe bowel ischaemia (a risk in left-sided surgery).

IVC and veins

- The IVC forms below/right of the aortic bifurcation and passes the diaphragm at **T8**. Veins vary more than arteries.
- The **left renal vein** crosses in front of the aorta below the SMA. It receives gonadal (right-angle entry), adrenal (with or without inferior phrenic) and ≥ 1 **lumbar vein** (avulsion risk). The **right** is short, without tributaries, duplicated in one-sixth. Both enter the IVC at L1.
- The **right gonadal vein** enters the anterolateral IVC (right renal vein in up to 10%). The **right adrenal vein** is short and enters the posterolateral IVC.
- Left varicocele predominates: longer vein, perpendicular entry. A **new right varicocele** needs imaging for a retroperitoneal mass, e.g. renal cell carcinoma with venous thrombus.
- Lumbar veins drain via ascending lumbar veins (behind psoas) to azygos/hemiazygos.
- Dissect the right ureter **lateral** to the gonadal vein; medially risks IVC injury and gonadal avulsion.
- The inferior mesenteric vein (**IMV**) lies in the left mesocolon, joins the superior mesenteric vein in 40%, and at reoperation is mistaken for the gonadal vein.

Variants (essential)

- P1 P3 IVC anomalous development and retrocaval ureter; Retrocaval ureter — *renal collar embryology, circumaortic/retroaortic vein, retrocaval ureter types and repair*
- † A retroaortic left renal vein (≈ 2–3%) is injured when lumbars are divided or the aorta rolled in retroperitoneal lymph node dissection (RPLND).
- † A circumaortic vein (≈ 5–7%) has a posterior limb that mimics a lumbar vein, so check the pre-operative computed tomography (CT).
- † Retrocaval ureter is a **venous** anomaly. It shows a "reverse-J" on pyelogram and is repaired by dismembered pyeloplasty in front of the IVC.

6 · Lymphatics

Why this matters: RPLND templates follow this predictable spread.

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *right-to-left cephalad flow, cisterna chyli L1–L2, three nodal regions, landing zones (right interaortocaval, left para-aortic), right-to-left crossover only*
- Lumbar nodes are the primary drainage of the lateral-branch organs: kidney, adrenal, ureter and gonad.

> **KEY POINT**
> The testis is embryologically retroperitoneal, so it drains to where it began. Crossover is one-way (right to left), so right templates include interaortocaval and some para-aortic tissue. *See Ch 80, 81*

7 · Nerves

Why this matters: ejaculatory loss and numbness come from these nerves.

Autonomic

- The **adrenal medulla** is a modified sympathetic ganglion (preganglionic fibres synapse directly).
- **Parasympathetic** outflow is craniosacral (III, VII, IX, X, S2–S4), with ganglia in the organ wall. **Sympathetic** outflow is thoracolumbar, **T1–L2**.

- **Sympathetic trunks** lie medial to psoas, behind the great vessels; lumbar vessels cross them at right angles, so they are met when lumbars are divided.
- The renal plexus (aorticorenal ganglion) supplies kidney, ureter and some testis. The superior hypogastric plexus runs from lower aorta to front of L5.
- **Ejaculation**: † postganglionic fibres from the **T12–L3** ganglia cross in front of the aorta and IVC and converge in the hypogastric plexus below the IMA. They drive **emission** and **bladder neck closure**, so extensive dissection causes anejaculation or retrograde ejaculation; erection (parasympathetic) is spared. *See Ch 81*

Lumbar (lumbosacral) plexus

Nerve	Roots	Course / motor	Sensory
Subcostal	T12	Below 12th rib	Skin over hip
Iliohypogastric	L1	Over quadratus; internal oblique, transversus	Gluteal, pubic
Ilioinguinal	L1	As above	Upper medial thigh, penile root, anterior scrotum
Genitofemoral	L1–L2	**On anterior psoas**; genital branch to cremaster	Anterior scrotum; upper anterior thigh
Lateral cutaneous thigh	L2–L3	Sensory only	Lateral thigh
Obturator	L2–L4	Medial to psoas, to obturator canal; adductors	Medial thigh
Femoral	L2–L4	Lateral to psoas; quadriceps	Anterior thigh, medial leg

Nerve	Setting	Result
Genitofemoral	Psoas hitch, laparoscopic varicocelectomy, RPLND on psoas	Anterior scrotal/thigh numbness
Obturator	Lateral-wall TURBT (bladder tumour resection); pelvic node dissection	Adductor jerk causing bladder perforation
Femoral	Retractor blade on psoas; lithotomy	Loss of knee extension
Hypogastric plexus	Extensive retroperitoneal dissection	Anejaculation/retrograde ejaculation

8 · Imaging anatomy

Why this matters: the pre-operative CT shows the variants that bleed.

- Axial CT from top down: upper poles, renal vessels (left renal vein in front of aorta), pelves with the **duodenum against the right hilum**, lower poles.
- † Before RPLND or nephrectomy, check accessory renal arteries, retro-/circumaortic left renal vein and nodal size by zone.

79 Neoplasms of the Testis

1 · Epidemiology and risk factors

Why this matters: cure is now expected, so the aim is to cure with the least toxicity.

- Over 95% are **germ cell tumours (GCTs)**: seminoma or non-seminomatous GCT (NSGCT); 95% testicular, 5% extragonadal. Commonest solid malignancy in men **20–40 y**, peaking at 25–35.
- Cisplatin raised metastatic cure from 5–10% to **80–90%**, so the aim is **minimum toxicity without losing curability**. Deaths come from platinum resistance and failure to clear residual disease.
- Bilateral in **2%**, mostly metachronous; histology discordant in 30–50%. Highest in Scandinavia; rising.
- **Five risk factors**: White race, cryptorchidism, family history, personal history, germ cell neoplasia in situ (GCNIS); also infertility.
  - Undescended testis (UDT): relative risk (RR) **4–6**, 2–3 if orchidopexy before puberty; contralateral descended testis ≈1.7.
  - An affected brother (RR 8–12) matters more than a father (2–4). A previous GCT gives a 12-fold contralateral risk, yet the 15-y cumulative incidence is only 2%.
- **GCNIS** is an arrested fetal gonocyte formed **before birth**. It lies beside invasive GCT in 80–90% and invades in **50% at 5 y** and 70% at 7 y. Contralateral GCNIS: 5–9%, ≈36% with atrophy, UDT or age <40.
- Microlithiasis alone needs no work-up; it matters only after a previous GCT.

> **KEY POINT**
> GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes risk.

2 · Pathology and classification

Why this matters: histology decides markers, spread and need for surgery.

- P3 Classify testicular tumors and describe management of seminoma — *WHO 2016 groups, i(12p), any NSGCT element = NSGCT, seminoma IHC*

Not in the paper answer:

- Isochromosome 12p, i(12p), proves GCT origin in a carcinoma of unknown primary and in somatic-type malignancy from teratoma.
- Expert pathology review changes the subtype in ≈1/3 and lymphovascular invasion (LVI) status in ≈1/5.

Subtype	Markers	IHC / histology	Hallmark
Seminoma	Human chorionic gonadotrophin (hCG) ≈15%; **never alpha-fetoprotein (AFP)**	CD117+, CD30−, placental alkaline phosphatase (PLAP)+	4th–5th decade; can transform to NSGCT
Spermatocytic	None	OCT3/4−, PLAP−; three cell types	6th decade; never mixed, not with UDT; **benign** unless sarcomatous/anaplastic

Subtype	Markers	IHC / histology	Hallmark
Embryonal carcinoma (EC)	AFP/hCG or normal	AE1/AE3+, OCT3/4+, CD30+†, c-KIT−	Totipotent; spreads with normal markers; predicts occult spread
Yolk sac	AFP, not hCG	Schiller–Duval bodies (≈50%), hyaline globules	Pure form: children, mediastinum
Choriocarcinoma	Very high hCG	Syncytio- and cytotrophoblasts	Spreads by blood (lung, liver, brain); bleeds; poor risk
Teratoma	Normal/mild AFP	≥2 germ layers; maturity irrelevant after puberty	Benign histology, malignant genetics; **chemoresistant**

IHC = immunohistochemistry.

- Teratoma is present in ≈half of mixed GCTs. Unresected, it causes growing teratoma, unresectability, somatic transformation (rhabdomyosarcoma, adenocarcinoma, primitive neuroectodermal tumour (PNET); late, at metastatic sites) or late relapse, so surgery is integral to cure.
- hCG shares its α-subunit with thyroid-stimulating hormone (TSH) and luteinising hormone (LH), so it can cause hyperthyroidism and gynaecomastia.

> **KEY POINT**
> Management follows the most chemoresistant element: EC drives occult metastasis, choriocarcinoma haemorrhage, teratoma surgery.

3 · Presentation

Why this matters: about a third are first misdiagnosed, and delay worsens stage.

- A **painless mass** is commonest. Metastases at diagnosis: ≈2/3 NSGCT, 15% seminoma. Symptoms from metastases in 10–20%: back pain, leg swelling, dyspnoea, left supraclavicular node. Gynaecomastia ≈2%.
- Re-examine epididymo-orchitis 2–4 weeks after antibiotics. ≈1/3 are first misdiagnosed. A man of 15–50 with a retroperitoneal or left supraclavicular mass needs testicular examination, ultrasound and markers.

4 · Diagnosis and initial management

Why this matters: inguinal orchidectomy is both the diagnosis and, for many, the cure.

Scrotal ultrasound (US)

- GCT is **hypoechoic**: homogeneous suggests seminoma; heterogeneous, cystic or calcified suggests NSGCT. Magnetic resonance imaging (MRI) if equivocal.
- In metastatic GCT with normal palpation, US seeks a **burned-out primary** (scar, calcification), which still needs orchidectomy. Normal testes mean extragonadal GCT.
- An impalpable <10 mm lesion with normal markers is GCT in 20–50%: orchidectomy, inguinal exploration with frozen section, or serial US.

Serum tumour markers

- P1 Tumour markers in urology — *sources, never-raised-in, half-lives, S stage, raised AFP = NSGCT*

Not in the paper answer:

- False positives: AFP with liver disease, hepatocellular or gastric cancer and ataxia telangiectasia; hCG with LH cross-reactivity in hypogonadism (normalises 48–72 h after testosterone) and marijuana; lactate dehydrogenase (LDH) with muscle injury and lymphoma.
- After orchidectomy, a slow decline with negative imaging is monitored, but **rising markers mean chemotherapy**.
- An unstable patient with very high AFP/hCG gets chemotherapy first and orchidectomy later, because the testis is a sanctuary.
- Half-lives: AFP **5–7 days**, hCG **24–36 h**, LDH 24 h.
- **Post-orchidectomy values** (within 4 weeks of treatment) set the S stage; markers never decide whether to remove the testis.
- miR-371a-3p is GCT-specific and is not raised by teratoma.

Radical inguinal orchidectomy

- Remove testis and cord to the internal ring within 1–2 weeks. **Trans-scrotal approach or biopsy is contraindicated** (inguinal/pelvic spread, local recurrence). Orchidectomy alone cures 80–85% of clinical stage (CS) I seminoma and 70–80% of CS I NSGCT.
- **Testis-sparing surgery**: synchronous bilateral or solitary testis (<2–3 cm, adequate androgen), or suspected benign lesion; frozen section and biopsies; GCNIS gets ≥20 Gy (deferrable for paternity).
- Contralateral biopsy if atrophy, UDT, age <40 or suspicious US. Extragonadal: raised markers with normal testes are diagnostic; otherwise biopsy for i(12p).

5 · Staging

Why this matters: stage and marker level choose between surveillance, surgery and chemotherapy.

Spread and imaging

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *retroperitoneum first site 70–80%, landing zones, crossover*

Not in the paper answer:

- Staging computed tomography (CT) also shows a retroaortic renal vein or lower-pole artery before retroperitoneal lymph node dissection (RPLND).
- Base decisions on CT and markers done within 4 weeks of treatment, because GCT grows fast.
- CT abdomen–pelvis for all misses disease in **25–35% of CS I NSGCT**, so landing-zone nodes of 5–9 mm are suspicious. Equivocal nodes with normal markers: repeat CT in 6–8 weeks.
- Chest X-ray for all; chest CT if markers raised, abdominal disease, abnormal X-ray, or CS I NSGCT with LVI/EC. Isolated mediastinal nodes need biopsy (lymphoma, sarcoidosis).
- Brain imaging if hCG >10,000 IU/L. No routine bone scan and no fluorodeoxyglucose positron emission tomography (FDG-PET) at diagnosis.
- Offer **sperm cryopreservation** before RPLND, chemotherapy or radiotherapy; up to half are oligospermic.

TNM staging — the only TNM with a serum (S) category

Category	Definition
pTis	GCNIS
pT1	Testis/epididymis, **no LVI**; may invade tunica albuginea
pT2	LVI, or tunica vaginalis

Category	Definition
pT3 / pT4	Spermatic cord / scrotum
N1 / N2 / N3	≤2 cm / >2–5 cm / >5 cm (pN1 ≤5 nodes, pN2 >5 or extranodal; pN invalid after chemotherapy)
M1a / M1b	Non-regional nodes or lung / other viscera
S1	LDH <1.5×N *and* hCG <5000 mIU/mL *and* AFP <1000 ng/mL
S2	LDH 1.5–10×N *or* hCG 5000–50,000 *or* AFP 1000–10,000
S3	LDH >10×N *or* hCG >50,000 *or* AFP >10,000

Stage	Criteria
IA / IB	pT1 / pT2–4, N0 M0 S0
IS	Any pT, N0 M0, **S1–3** (raised markers, no visible disease)
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	M1a, S0–1
IIIB	N1–3 or M1a with S2
IIIC	N1–3 or M1a with S3, or **any M1b**

IGCCCG risk — metastatic disease

Risk	NSGCT	Seminoma
Good	Testis/retroperitoneal primary, no NPVM, AFP <1000, hCG <5000, LDH <1.5×N (56%; 5-y survival 92%)	Any primary, no NPVM, any hCG/LDH, normal AFP (90%; 86%)
Intermediate	Any of AFP 1000–10,000, hCG 5000–50,000, LDH 1.5–10×N (28%; 80%)	**NPVM** (10%; 72%)
Poor	**Mediastinal primary**, NPVM, AFP >10,000, hCG >50,000 or LDH >10×N (16%; 48%, now ≈71%)	**None**

IGCCCG = International Germ Cell Cancer Collaborative Group; NPVM = non-pulmonary visceral metastases.

> **KEY POINT**
> IGCCCG uses markers at the start of chemotherapy, not pre-orchidectomy values, and is invalid at relapse. Seminoma risk depends on NPVM alone; mediastinal NSGCT is always poor risk.

6 · Treatment principles

Why this matters: seminoma and NSGCT follow different pathways.

- Centralise care, give full-dose cisplatin despite cytopenias or renal impairment, and resect all residual disease.

	Seminoma	NSGCT
CS I / II / III	85 / 10 / 5%	≈1/3 each
Occult metastasis CS I	10–15%	25–35%
Radiosensitive	Exquisitely	No (except brain)
Metastatic teratoma	Rare	Frequent, so surgery needed

	Seminoma	NSGCT
Markers	Rarely useful	Central

- Seminoma relapses contain NSGCT in 10–15%, so biopsy non-responders. Expert nerve-sparing RPLND has the best long-term toxicity profile.

GCNIS management

- Options: orchidectomy, radiotherapy **≥20 Gy** (<20 Gy allows recurrence) or observation. Radiotherapy spares Leydig cells, yet up to 40% later need testosterone.
- Subfertile men: serial US or sperm retrieval first. Normal contralateral testis with paternity wish: orchidectomy (scatter harms it). Cisplatin only partly clears GCNIS, so biopsy ≥2 y later or give radiotherapy.

7 · NSGCT management

Why this matters: every CS I option cures ≈100%, so the choice is about overtreatment and toxicity.

CS I NSGCT

- Occult metastasis is present in 20–30%, so adjuvant therapy overtreats 70–80%.
- **Risk factors** are **LVI** (strongest) and **EC predominance**. High-risk tumours relapse in ≈50%, low-risk in <20%.
- **Surveillance**: relapse ≈28%, >90% within 2 y, 65–75% retroperitoneal; chemo-naive and curable.
- **Primary RPLND** removes teratoma (15–25% of occult nodes). Pathological stage II in 19–28%, mostly cured by RPLND alone. Retroperitoneal recurrence <2%, so no routine abdominal CT. pN1: observe; pN2–3 (relapse >50%) gets **2 cycles adjuvant** (relapse ≤1%) or observation.
- **Bleomycin–etoposide–cisplatin (BEP) ×1** gives the lowest relapse (≈2–3%) but leaves teratoma, exposes all to late toxicity, and relapses are chemoresistant.
- Template: full bilateral nerve-sparing (ejaculation >90%) is standard and mandatory for suspicious nodes; a modified template is acceptable for clinically negative nodes.

ALGORITHM

1. Assess CS I NSGCT for LVI and EC predominance.
2. If low risk (neither), then surveillance.
3. If high risk, then surveillance, BEP×1 or expert nerve-sparing RPLND, chosen by preference and toxicity.

KEY POINT

Surveillance maximises quality-adjusted survival when relapse risk is <≈1/3. All options cure ≈100%; they differ in overtreatment versus difficulty of salvage.

CS IS, IIA–B

- **CS IS** is treated with induction chemotherapy.
- **CS IIA–B** with raised markers or nodes **>3 cm** gets chemotherapy, with or without post-chemotherapy RPLND (PC-RPLND). With normal markers and nodes <3 cm, **primary RPLND** is used: 13–35% are pN0, ≈30% have teratoma, and half avoid chemotherapy.

CS IIC–III: induction by IGCCCG

- P2 CRE & BEP — *BEP×3/EP×4 good, BEP×4 (VIP if lungs poor) intermediate/poor, toxicities*

Not in the paper answer:

- BEP×3 equals BEP×4; etoposide–cisplatin (EP) ×3 and carboplatin are inferior.
- Second line is TIP×4 (paclitaxel, ifosfamide, cisplatin), VeIP×4 (vinblastine) or VIP; TIP is best for late relapse.
- High-dose chemotherapy (HDCT: carboplatin–etoposide with stem cells) failed in three randomised trials (two first-line), so it is for specialised centres. Carboplatin replaces cisplatin because nephro- and neurotoxicity limit escalation.
- Ifosfamide causes haematological and urological toxicity. Slow marker decline is adverse; intensification improved progression-free but not overall survival, so BEP×4 stays standard.

Post-chemotherapy residual masses (NSGCT)

- With normal markers and a residual mass **>1 cm**, do **PC-RPLND** with a full bilateral template. Histology: necrosis ≈40%, teratoma ≈45%, viable tumour ≈15%.
- No model, imaging or **FDG-PET** predicts necrosis (teratoma is not avid; ≈20% false negative), so always resect.
- Complete response (≤1 cm): observe, or resect if pre-chemotherapy >3 cm (controversial).
- Multiple sites: resect all, retroperitoneum first (its necrosis predicts necrosis elsewhere). Somatic transformation (6–8%) needs complete resection.
- After viable tumour, adverse factors are intermediate/poor risk, incomplete resection and >10% viable cells. Inadequate post-chemotherapy surgery (PCS) caused most deaths in the French BEP×3 vs EP×4 trial.
- Marker-positive partial response or progression: second-line chemotherapy.

Relapsed NSGCT

- P1 Growing teratoma syndrome — *enlarging mass, normal markers: stop chemotherapy, resect*
- **Chemo-naive relapse**: induction by IGCCCG, cure >95%; small marker-negative retroperitoneal relapse may have RPLND.
- **Early relapse (≤2 y) after chemotherapy** gets TIP/VeIP/VIP×4 or HDCT; residual masses then need post-salvage surgery.
- **Desperation surgery** is for rising markers with a single resectable site; survival is 33–57%.
- **Late relapse (>2 y)** occurs in ≈3%. Yolk sac tumour is commonest and chemoresistant, so **complete resection** is the treatment, with TIP if needed; biopsy if markers are normal.

8 · Seminoma management

Why this matters: most seminomas are CS I and are cured whichever option is chosen.

- P3 P2 Classify testicular tumors and describe management of seminoma; Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis. — *CS I options table, risk factors, Calvert, stage II–III, residual mass, follow-up*

Not in the paper answer:

- Radiotherapy causes scatter oligospermia in 8% and second malignant neoplasm (SMN) in ≈18% at 25 y. Pelvic nodes are involved in only 1–3% without prior surgery, which is why the field is para-aortic.
- Relapse after carboplatin is treated as chemo-naive. On surveillance hCG detects <5% of relapses, examination and X-ray almost none.
- Primary RPLND for IIA–B is supported by SEMS and COTRIMS (≈10% relapse).
- Viable tumour in a residual mass: >3 cm 13–55%, <3 cm 0–4%. Complete resection is achieved in 58–74%.

- CS I is ≈80%. Risk factors are tumour **>4 cm** and rete testis invasion (relapse 12/16/32% with 0/1/2), but they are not validated, so surveillance is preferred for all.
- Use a dog-leg field if there was prior scrotal or inguinal surgery. Carboplatin ×2 gives relapse ≈1.5%.
- Seminoma surveillance is harder than NSGCT: median relapse is at 14 months (NSGCT 4–8).
- IIA–B nonbulky (≤3 cm): dog-leg (DL) radiotherapy 25–30 Gy with a 5–10 Gy boost, BEP×3/EP×4, or primary RPLND. Bulky or multiple nodes get chemotherapy.
- Residual masses are necrosis in ≈90%, and many regress over 12–18 months.

ALGORITHM

1. If the residual seminoma mass is <3 cm, then observe.
2. If >3 cm, then FDG-PET ≥6 weeks after chemotherapy (earlier scans give false positives).
3. If PET-negative, then observe.
4. If PET-positive, then template PC-RPLND (or repeat PET/observe). Mass-only excision gives 33% local relapse.

- Relapse after radiotherapy gets first-line chemotherapy; after chemotherapy, VeIP×4 or HDCT. With normal markers, biopsy for transformed NSGCT. Late seminoma relapse may be favourable.

9 · Brain metastases

Why this matters: they can bleed fatally.

- Present in ≈1% at diagnosis, with **choriocarcinoma or very high hCG**. Give BEP×4, then resect residuals; use radiosurgery if unresectable. Brain relapse after complete response is worse.

10 · Treatment sequelae and follow-up

Why this matters: survivors live decades with late effects.

- RPLND can cause ejaculatory loss (>80% after full template without nerve sparing) and small-bowel obstruction. Orchidectomy alone causes hypogonadism in 10–20%.
- Chemotherapy toxic death: 0–2.4% (good), 3–4.4% (intermediate/poor). Azoospermia is near universal; ≈half recover by 2 y, 80% by 5 y.
- Late effects: Raynaud 20–45%, neuropathy 14–43%, ototoxicity, hypogonadism 15–40%, **cardiovascular disease** and **SMN** (60–100% excess after cisplatin or radiotherapy, 200% after both).
- Follow-up is most intensive in years 1–2 (NSGCT) or 1–3 (seminoma); seminoma surveillance beyond 5 y; after carboplatin ≥5 y; after full RPLND, chest imaging and markers. Survivorship care†: blood pressure, lipids, glucose, testosterone.

KEY POINT

Because cure is expected, late cardiovascular disease and second cancers, which rise with each added modality, decide between equally curative options.

11 · Non-germ-cell tumours†

Why this matters: they need different treatment.

- †**Leydig cell**: the commonest sex cord–stromal tumour (≈1–3%). It causes precocious puberty in children and gynaecomastia in adults. Reinke crystals; inhibin, calretinin and Melan-A positive. ≈10% are malignant (>5 cm, necrosis, LVI, mitoses); these resist chemo- and radiotherapy, so nodes need RPLND. A small marker-negative endocrine lesion suits testis-sparing excision.
- †**Sertoli**: mostly benign. The large-cell calcifying variant occurs with Carney complex and Peutz–Jeghers (bilateral, gynaecomastia).
- †**Lymphoma**: the commonest testicular tumour in men **>50–60 y** and the commonest bilateral one. Diffuse large B-cell lymphoma is treated by orchidectomy, then R-CHOP with central nervous system prophylaxis and contralateral testis radiotherapy.

12 · Landmark trials

Why this matters: these set current standards.

Trial	Design	Result	Take-home
IGCCCG 1997	Pooled 5202 NSGCT, 660 seminoma	Markers, NPVM, mediastinal primary predict outcome	Risk groups select regimen
PVB vs BEP	RCT, advanced	Equal survival, less neurotoxicity	BEP standard
BEP×3 vs ×4	RCT, good risk	Equal survival	BEP×3
VIP vs BEP×4	RCT, intermediate/poor	Equal, more toxicity	VIP only if lungs at risk
MRC TE19/EORTC 30982	Carboplatin ×1 vs radiotherapy, CS I seminoma	5-y RFS 94.7% vs 96%	Equivalent
MRC 20 vs 30 Gy	RCT, CS I seminoma	Equal relapse, less toxicity	20 Gy PA
Albers (German)	BEP×1 vs unilateral RPLND, CS I NSGCT	2-y PFS 99% vs 92%	Favours BEP×1
Williams	Adjuvant ×2 vs observation, pathological stage II	Relapse 6% vs 49%; same survival	Both acceptable
MRC surveillance	2 vs 5 CTs, CS I NSGCT	No difference	Fewer scans safe

RCT = randomised controlled trial; RFS = relapse-free survival; PFS = progression-free survival; PA = para-aortic; MRC = Medical Research Council.

80 Surgery of Testicular Tumours

1 · Evaluation of the testis mass

Why this matters: it is urgent, because delay escalates the intensity of curative therapy.

- History, then examination (supraclavicular, breasts, abdomen, scrotum). Scrotal ultrasound (US) mainly checks the **contralateral testis** (bilateral synchronous ≈ **1%**). Markers, alpha-fetoprotein (AFP), β-human chorionic gonadotrophin (β-hCG) and lactate dehydrogenase (LDH), give diagnosis, stage and post-orchidectomy baseline.
- Discuss a prosthesis; † offer sperm banking before orchidectomy.

2 · Radical inguinal orchidectomy

Why this matters: the right first operation avoids recurrence.

Treatment of choice: **early cord control** before handling the testis; testis, epididymis and cord to the internal ring are removed. † A scrotal approach breaches a second lymphatic field (inguinal), which leads to local recurrence.

ALGORITHM

1. Make a 3–5 cm incision over the canal and open the external oblique.
2. **Preserve the ilioinguinal nerve**.
3. Mobilise the cord at the pubic tubercle and apply a **Penrose early clamp**.
4. Deliver the testis and divide the gubernaculum.
5. Dissect the cord to the internal ring.

- **High ligation**: ligate vas and gonadal vessels **separately** with **non-absorbable** suture, leaving a **1–2 cm** tail. This marks the stump for excision at retroperitoneal lymph node dissection (RPLND); the vas can stay.
- Use scrotal support for 2–3 days. † A prosthesis can go in at the same sitting.

KEY POINT

† **Scrotal violation**: inguinal exploration, high cord ligation, excise the scrotal scar. Routine hemiscrotectomy is unnecessary; chemotherapy-era outcomes are not compromised.

Partial orchidectomy (testis-sparing surgery)

- P4 Testis sparing surgeries for CA testis — *indications (solitary/bilateral, small, polar), intraoperative US, cold ischaemia <30 min, frozen section, GCNIS 18–20 Gy, recurrence: completion orchidectomy*

Not in the paper answer:

- Incise the tunica vertically for ventral midline and horizontally for medial/lateral lesions (spares segmental arteries); three-point fixation.
- Observation of germ cell neoplasia in situ (GCNIS) gives local recurrence of ≈ 7.5%.
- Masses **<5 mm** with negative markers: low malignancy, so selective surveillance is an option.
- Needs **all three**: polar, **≤2 cm**, contralateral testis compromised/absent. **Never elective with a normal contralateral testis.**
- Cold ischaemia can be omitted if resection takes **<30 min**.
- GCNIS options: completion orchidectomy, **18–20 Gy** or observation. **Local recurrence always needs completion orchidectomy**.

Delayed orchidectomy

Life-threatening metastatic germ cell tumour (GCT) gets chemotherapy first, then **delayed orchidectomy in all**, because the testis is a sanctuary with **discordant response**; it usually shows teratoma or necrosis. An apparent extragonadal retroperitoneal primary shows intratubular neoplasia in up to 42%; lateralising nodes mean orchidectomy that side. Stage with computed tomography (CT) (or magnetic resonance imaging), no PET; repeat markers.

3 · RPLND — terminology and indications

Why this matters: the setting predicts difficulty.

Term	Indication
Primary RPLND	High-risk clinical stage (CS) I or low-volume stage II non-seminomatous GCT (NSGCT), **normal markers**
Post-chemotherapy (PC-RPLND)	After induction; residual **>1 cm**, normal markers
Salvage PC-RPLND	After second-line chemotherapy
Desperation	Markers still elevated; resectable
Redo	Retroperitoneal recurrence after prior RPLND
Late relapse	**>24 months** after complete response

Each step down means more chemoresistant disease, harder surgery.

Preoperative planning

- CT within **6 weeks**, markers within **7–10 days**; note vascular variants; no bowel prep.
- After **bleomycin**: lung function tests and alert anaesthesia to use **low FiO_2, restrictive fluids**.
- Inferior vena cava (IVC) thrombus: total occlusion needs IVC resection; partial needs reconstruction.

4 · Templates

Why this matters: it balances clearance against ejaculation.

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *landing zones, bilateral infrahilar and modified templates, omit suprahilar/interiliac, 3–23% out-of-template, PC-RPLND full bilateral; modified only if ≤5 cm in landing zone + normal markers + good/intermediate risk; L1–L4 fibres to hypogastric plexus*
- Bilateral template limits: renal vessels, ureters, common iliac arteries. No randomised trial of extent. A modified post-chemotherapy template needs a lesion **≤5 cm**.

5 · Open RPLND technique

Why this matters: completeness depends on technique.

- Approach is **transperitoneal** or **extraperitoneal** (bowel contained, so less ileus).
- Incise from the caecum to the inferior mesenteric vein (IMV), and along the line of Toldt to the foramen of Winslow. Free the duodenum and pancreas. Use **liberal clips** to prevent chyle leak.
- **Split and roll**: split at 12 o'clock on the aorta from below the left renal vein down to the inferior mesenteric artery (IMA); ligate cephalad lymphatics; roll packets off; ligate lumbars.

- Lumbar arteries: usually 3 pairs, **2nd pair at IMA level**. The **common lumbar vein** is largest. The **right sympathetic trunk**, posterolateral to the IVC, is mistaken for lymphatic tissue. Spare the genitofemoral nerve.
- Excise the gonadal vein to the internal ring and orchidectomy suture.

Nerve sparing

- Left fibres run along the lateral aortic border, under the left renal vein and around the left common iliac, near the gonadal artery and IMA. Right fibres emerge **beneath the IVC** and cross the aorta obliquely, so they are at risk when the superior or common lumbar veins are ligated.

> **KEY POINT**
> Identify fibres **prospectively, before dividing anything**; loop them and pass the packet beneath. Avoid traction, which causes neuropraxia and transient anejaculation.

Postoperative care

- Optionally close the mesenteric root (volvulus). No routine drain; if one is used, remove it at **<100 mL/24 h** on diet. No nasogastric (NG) tube; low-fat diet on days 1–2; mobilise on day 1; home on days 3–5.

6 · Auxiliary procedures (PC-RPLND 23–45%)

Why this matters: extra procedures must be planned.

- **Nephrectomy** is commonest, usually **left**. The strongest predictor is a residual mass **>10 cm**. Risk rises from fibrosis to teratoma to viable tumour, and in salvage, desperation and redo cases. It may affect future cisplatin eligibility.
- **Vascular** repair is planned in every PC-RPLND (subadventitial aortic injury bleeds fatally). Repair with a patch or a PTFE/Dacron graft.
- **IVC**: narrowing **<25%** is repaired by venorrhaphy, >25% by patch or graft. Ligation is acceptable (collaterals); grafts thrombose. With a mass >5 cm, ≈20% need IVC work. **The aorta is never ligated**; duodenal injury with an aortic graft risks aortoduodenal fistula.
- **Liver**: histology is discordant in ≈half; resect, or use microwave ablation if **<3 cm** and unresectable. **Thorax**: 25–50% discordant, so do RPLND first, then resect lesions >1 cm unless the retroperitoneum shows fibrosis only (which predicts thoracic necrosis in ≈89%). **Pelvis**: mostly teratoma. **Retrocrural**: approach transdiaphragmatically along the right crus.

7 · Decision making

Why this matters: who is observed and who is treated.

- **Complete remission (≈70%)**: generally observe, though ≤1 cm residuals hold teratoma in ≈ 1/5 and viable tumour in ≈ 1/20.
- **Necrosis is predicted by** no teratoma in the primary, normal pre-chemotherapy AFP/β-hCG, raised LDH, >50% shrinkage and residual <2 cm.
- **After primary RPLND**: pN0 or teratoma only, observe; pN1, observe; pN2, 2 cycles or observation. Observation cures ≈70%; relapses need ≥1 extra BEP (bleomycin, etoposide, cisplatin) or 2 extra EP cycles.

8 · Histology of residual mass after chemotherapy

Why this matters: histology decides further treatment.

Finding	Frequency	Outcome/action
Fibrosis/necrosis	**≈40%**	Relapse-free (RFS) and cancer-specific survival (CSS) ≈95%; no adjuvant
Teratoma	**≈45%**	RFS 80–90%; must be resected
Viable GCT	**≈15%**	Survival 50–70%; consider 2 adjuvant cycles

KEY POINT

Resect every NSGCT residual >1 cm: no imaging (including PET †) or marker separates necrosis from teratoma, and teratoma is chemoresistant, grows, relapses late and transforms. Surgery is both staging and the only treatment; histology decides further therapy.

- **Fizazi** model for viable GCT: incomplete resection, ≥10% viable cells and IGCCCG intermediate/poor risk. With 0, 1 or ≥2 factors, 5-y overall survival (OS) is 100/83/51%.
- Adjuvant chemotherapy: NCCN for incomplete resection or intermediate/poor risk; EAU for incomplete resection only. Not after salvage chemotherapy.

9 · High-risk PC-RPLND

Why this matters: surgery can cure chemoresistant disease.

- **Salvage**: viable GCT is at least doubled; OS 60–75%. After high-dose chemotherapy (HDCT) viable tumour is ≈40%, yet surgery is still worthwhile.
- **Desperation**: OS ≈54%. Operate if confined, completely resectable, and markers declining/plateaued, or chemotherapy exhausted. Do **not** operate with rapidly rising markers or multisite progression. A plateau predicts >75% fibrosis or teratoma. **Postoperative marker normalisation** is the only OS predictor.
- **Redo** is mostly for technical failure (retroaortic/retrocaval tissue, lumbars), so do **complete great-vessel mobilisation the first time**.
- **Late relapse** (yolk sac, somatic-type malignancy) is treated by **surgery first**.

10 · Unique situations

Why this matters: NSGCT rules do not apply.

- **Seminoma residual**: dense desmoplasia, no teratoma, viable ≈10%. If **<3 cm**, observe; if **>3 cm**, do FDG-PET (≥6 weeks †). If avid, choose selected PC-RPLND, salvage chemotherapy or HDCT.
- **Sex cord stromal**: malignant features are >4 cm, necrosis, mitoses, atypia, infiltration and lymphovascular invasion. RPLND is of unproven benefit. Leydig tumours are almost always stage I; Sertoli tumours are more aggressive.

11 · Outcomes, fertility and complications

Why this matters: counsel on these before RPLND.

- RPLND spares spermatogenesis; 40–60% already have abnormal semen at diagnosis.
- **Ejaculation**: template or nerve sparing preserves antegrade ejaculation in **>90%** (primary) at no oncological cost. † If lost: pseudoephedrine/imipramine, post-ejaculatory urine sperm, electroejaculation, testicular sperm extraction.
- Complications: primary 10.6–24%, PC up to 56%. Mortality: primary ≈0, PC <1%.

Complication	Key facts / management
Pulmonary (bleomycin)	PC 3–5%, acute respiratory distress syndrome. Low FiO_2, restrictive fluids; tachycardia alone is no fluid trigger; accept SpO_2 **88–90%**
Ileus	Up to 26% (commonest); extraperitoneal approach, no NG tube, early feeding
Chylous ascites	Primary 0.2–2.1%, PC 2–12%; linked to suprahilar dissection (cisterna chyli L1–L2)
Lymphocele	Symptomatic ≤1.7%; air or rim enhancement means infection: drain, with or without antibiotics
Venous thromboembolism (VTE)	Primary <1%, PC 0.1–3.1%; drug prophylaxis if prior chemotherapy, VTE or obesity
Neurological	Femoral neuropraxia; paraplegia with bulky disease

- P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — *milky tap, triglyceride 2–8× serum, protein >3 g/dL; stepwise management*

ALGORITHM

1. Paracentesis; drain if it reaccumulates.
2. Very-low-fat, high-protein diet with medium-chain triglycerides (MCT) and octreotide.
3. If no response, nil by mouth with total parenteral nutrition (TPN).
4. If refractory, lymphangiography with or without embolisation, peritoneovenous shunt or exploration.
5. Continue a low-fat diet ≥ **4 weeks** after resolution.

12 · Landmark trials

Why this matters: adjuvant therapy and PET use.

Trial	Design	Result	Take-home
Williams (Testicular Intergroup)	RCT, pathological stage II: 2 adjuvant cycles vs observation	Adjuvant near-abolished relapse; equal survival	Observe pN1; individualise pN2
SEMPET	FDG-PET in post-chemotherapy seminoma residuals	>3 cm: specificity 100%, NPV 96%; ≤3 cm PPV 37%	PET for >3 cm only

81 Minimally Invasive Retroperitoneal Lymph Node Dissection

- P3 P4 Minimal invasive retroperitoneal lymph node dissection; Robotic RPLND — *indications, templates, nerve sparing, robotic steps, lap vs robot, outcomes, AUA/EAU positions, miR-371a-3p*

Not in the paper answer:

- The National Comprehensive Cancer Network (NCCN) is silent on it.
- Benchmark recurrence-free ≈**85%** for limited pN+ without adjuvant chemotherapy; low MI recurrence is **confounded** by high pN0 and adjuvant rates.

1 · Indications and rationale

Why this matters: cure is near certain, so morbidity counts.

- Goals in early-stage disease: preserve cure, avoid **dual therapy** (surgery and chemotherapy) and minimise lifelong toxicity.
- **Primary** RPLND (retroperitoneal lymph node dissection) is for marker-negative clinical stage (CS) I non-seminomatous germ cell tumour (NSGCT), and small-volume stage II NSGCT and seminoma. It stages and cures occult disease, and spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0.
- **Post-chemotherapy (PC-RPLND)** is for NSGCT residual **>1 cm** and select seminomas.
- Minimally invasive RPLND (MI-RPLND) is the latest de-escalation, after dropping suprahilar dissection, nerve-sparing and modified templates. The robot adds fourth-arm retraction and **circumferential great-vessel dissection with lumbar control**.

> **KEY POINT**
> Much recovery gain came from Enhanced Recovery After Surgery (ERAS) pathways, which help every approach, so the benefit of access alone is smaller than it looks.

2 · Guidelines and controversy

Why this matters: evidence is weak.

- American Urological Association (AUA, early-stage): **may be offered** by a surgeon expert in germ cell tumour and minimally invasive surgery (MIS); patient told long-term data lack. European Association of Urology (EAU): **specialist centres** expert in open and MIS only.
- No noninferiority evidence, so it is the most controversial operation in urological oncology.
- Concerns: staging-only dissection (no lumbar division, low yields, excess adjuvant chemotherapy even in pIIA, so dual therapy; residual teratoma causing late relapse); wrong patients (stage IA operated; pN0 up to 83%); great-vessel surgery from a console by surgeons lacking open skills, with conversion **≈5%**; no series > 5 years, and peritoneal carcinomatosis reported.

3 · Patient selection

Why this matters: selection is the main safety tool.

- No absolute contraindications; relative: extensive prior surgery, morbid obesity.
- **Suited to MIS**: chemo-naive high-risk IB; small-volume II (1–2 landing-zone nodes, none **>3 cm**); PC solitary landing-zone mass with a small response (suggests teratoma) and little desmoplasia.

- **Favour open**: PC-RPLND in general, a large mass, large shrinkage from pre- to post-chemotherapy (desmoplasia), possible vessel reconstruction, somatic malignancy or retrocrural disease.

4 · Unchanging operative principles

Why this matters: access changes, the cancer operation must not.

- Imaging within **≈4 weeks**, markers within **1–2 weeks**.
- **MI-RPLND must completely duplicate open RPLND**: template, completeness and nerve-sparing are unchanged.

ALGORITHM

1. Safe entry and exposure.
2. **Split-and-roll** nodal removal.
3. Prospective nerve sparing.
4. **Lumbar vessel division**.
5. Spermatic cord removal, then closure.

- Chemo-naive: extended unilateral nerve-sparing template (debated); if enlarged nodes are found, convert to bilateral. Post-chemotherapy: **bilateral full template** is standard.
- Incomplete RPLND or mass-only excision worsens survival.

5 · Preparation, positioning and ports

Why this matters: bleeding can be sudden.

- Haemorrhage plan; after bleomycin, careful fluids and oxygen; **open set in theatre**. A high-fat meal the day before highlights lymphatics; no bowel prep. Pad pressure points (rhabdomyolysis).
- Lateral decubitus (≈45°) is familiar; bilateral needs repositioning. Laparoscopic supine: four 12-mm midline trocars. Robotic steep Trendelenburg: camera 4–6 cm below umbilicus, any template, with a **peritoneal hammock** (posterior peritoneum incised from Treitz round the caecum, edges sutured to anterior wall).

6 · Operative steps

Why this matters: each side has vascular traps.

- **Right**: Toldt line to foramen of Winslow, **Kocherise**; divide gonadal vein at the inferior vena cava (IVC); split over IVC from iliac crossing to left renal vein; clip retropancreatic lymphatics; roll paracaval packet, spare fibres, divide lumbars (Weck clips vessel side). **Do not ligate the right renal artery** in the interaortocaval space; preserve accessory lower-pole arteries.
- **Left**: divide gonadal vein at left renal vein; control the **lumbar vein entering the left renal vein**. The inferior mesenteric artery (IMA) may be ligated for mesocolic rotation.
- Lower pneumoperitoneum to check haemostasis; bag packets; **no routine drain**; discharge at **1–2 days**.

7 · Complications and management

Why this matters: vascular injury is most feared.

ALGORITHM

1. Major vascular injury: apply pressure and raise the pneumoperitoneum.
2. Inform anaesthesia, get blood and call vascular help.
3. If uncontrolled, keep pressure, undock and convert.

- For MIS repair, **identify the injury before acting: no blind clipping**.
- Chylous ascites: liberal clips at left renal vein and inferior mesenteric vein crossing (energy devices unreliable); rates match open. † Treat by low-fat/medium-chain triglyceride diet, then paracentesis, octreotide, parenteral nutrition, then ligation.
- Antegrade ejaculation is kept in **≥90%** after primary nerve-sparing.

8 · Outcomes vs open

Why this matters: equivalence is unproven.

- **No randomised trial**. Shorter stay, less blood loss, longer operative time; brisk bleeding is the unique risk.
- Robotic node yields approach the open benchmark of **≈30**, but adjuvant chemotherapy is still too frequent.
- PC data are too small.

9 · Landmark trials

Why this matters: completeness affects survival.

Trial	Design	Result	Take-home
GETUG T93BP	Good-risk NSGCT: BEP×3 vs EP×4	Most relapses/deaths without or with incomplete RPLND	Complete RPLND matters

82 Tumours of the Penis

1 · Overview

Why this matters: the primary and the groin nodes decide survival.

- Squamous cell carcinoma (SCC) is **~95%** of penile malignancy, and **0.4–0.6%** of male cancers in North America and Europe, but up to **10%** in parts of Asia, Africa and South America.
- Mean age is mid-50s; a meaningful minority are under 40. Prognosis rests on **primary stage and histology** and on **nodal metastasis**.

2 · Premalignant lesions: penile intraepithelial neoplasia (PeIN)

Why this matters: PeIN is curable with the organ kept.

- P1 Premalignant conditions in urology (including premalignant lesions of penis) — *PeIN definition, WHO 2022 all high grade, HPV-associated vs differentiated, E6/E7/p16, EQ/Bowen/Bowenoid table, LS 2.3–9%, Buschke–Löwenstein, organ-preserving treatment*

Not in the paper answer:

- Differentiated PeIN shows hyperplastic epithelium, keratin pearls, prominent intercellular bridges and atypical basal cells.
- Potassium titanyl phosphate (KTP) laser is an option; resurfacing has very low recurrence.
- Erythroplasia of Queyrat (EQ) is a red velvety glans lesion in uncircumcised men, mimicking balanitis; it progresses in **10–33%**.
- Bowen disease is a scaly shaft plaque mimicking psoriasis, eczema or superficial basal cell carcinoma (BCC); it progresses in **~5%**.
- Bowenoid papulosis: shaft papules in young men, linked to human papillomavirus (HPV); **does not progress**.
- Eponyms describe only site and look. EQ differs from balanitis by full-thickness atypia, bulbous rete, capillary proliferation and plasma cells. Metastasis from PeIN is extremely rare.
- Differentiated PeIN mimics pseudoepitheliomatous hyperplasia. Imiquimod is the topical choice for condyloma.

3 · Aetiology and risk factors of SCC

Why this matters: most risk factors are preventable.

Factor	Key point
No neonatal circumcision	Neonatal circumcision virtually eliminates invasive cancer; it protects against invasive SCC, **not CIS** (carcinoma in situ). Adult circumcision gives little or no protection
Phimosis	Present in **25–75%**; **7–10×** risk. It mediates the non-circumcision risk: smegma causes inflammation, which leads to fibrosis
HPV	Low risk 6, 11; high risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest**. Present in **~50%**; wives have ~3× cervical cancer risk
Human immunodeficiency virus (HIV)	~4× risk of penile cancer death; PeIN progresses ~6 years earlier
Tobacco	All forms; dose-dependent and independent of phimosis

Factor	Key point
Lichen sclerosus (LS)	Cancer in **2.3–9%**; HPV-**independent** SCC
Trauma, obesity	Scarred shaft; obesity causes buried penis and functional phimosis
† **Psoralen and ultraviolet A (PUVA)**	† Genital exposure markedly increases SCC, so shield the genitalia

- HPV-independent pathway: CDKN2A hypermethylation, TP53 mutation, MDM2. Inflammation acts through reactive oxygen species (ROS) and COX-2.
- **Not linked**: occupation, gonorrhoea, syphilis, herpes, alcohol, marijuana.
- **Prevention**: neonatal circumcision (AAP 2012: benefits outweigh risks), hygiene, no tobacco, HPV vaccination (nonavalent adds 31/33/45/52/58).

> **KEY POINT**
> Two routes converge on the preputial sac: HPV-driven (E6/E7, p16-positive, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision prevents almost none.

4 · Natural history and spread

Why this matters: spread reaches the groin long before distant sites.

- Sites: glans (~½), prepuce, coronal sulcus; shaft <2%. **Flat ulcerative tumours metastasise earlier**.
- **Buck fascia** is a temporary barrier; once breached with the tunica, tumour enters the corpora and spreads by blood.
- Distant metastases are uncommon (**1–10%**) and late. Untreated, death comes within **2 years** (femoral erosion haemorrhage, sepsis); there is no spontaneous remission.
- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *penile superficial/deep lymphatics, stepwise bilateral spread, no skips*

Not in the paper answer:

- Pelvic stations: external iliac, internal iliac, obturator. Distant sites: lung, liver, bone, brain; rare without nodal disease.

5 · Presentation and diagnosis

Why this matters: a deep biopsy sets every later decision.

- Induration, warty growth or rolled-edge ulcer, possibly hidden by phimosis; pain is slight. 15–50% delay >1 year.
- Examine size, fixation, corpora, base, scrotum and perineum, and **palpate both groins**.
- **Biopsy is mandatory before definitive therapy**: depth and grade predict occult nodes; superficial biopsies mislead. Use a dorsal slit, or frozen section then immediate resection.
- Labs: anaemia, leucocytosis, hypoalbuminaemia. **Hypercalcaemia without bone metastases** comes from PTH-related substances, tracks bulky inguinal disease and may resolve after node excision. Treat with saline, then bisphosphonate, then calcitonin; diuretic only if overloaded.
- Differential: condyloma, BXO, chancre, chancroid, herpes, LGV, granuloma inguinale, TB.

6 · Histology

Why this matters: subtype and grade predict occult nodes.

Subtype (WHO 2022)	HPV	Features	Nodal risk
Usual (keratinising) **45–65%**	Independent	Usually grade II; lymphovascular (LVI) or perineural invasion (PNI) in ~⅓	25–40%
Warty	Associated	Cauliflower, fibrovascular cores; slow	17–25%
Basaloid	Associated	Small cells, necrosis, deep	**>½ node positive**
Papillary	Independent	Low grade, no koilocytosis	≤12%
Verrucous	Independent	**Broad pushing border**	**No metastasis**; recurs if under-excised
Sarcomatoid	Independent	Spindle cells	**Most aggressive**
Rare: pseudoglandular, adenosquamous; clear cell, lymphoepithelioma-like (HPV)	—	Deep invasion	Pseudoglandular, clear cell worst

- **Any anaplastic cells make it grade III** (half are heterogeneous). **LVI** predicts nodes in every study; **PNI** strongly.
- **High-risk HPV (hrHPV) or p16 positivity means better disease-specific survival (DSS)**, not better nodal status. Adverse markers are p53, PD-L1, CRP and neutrophil-to-lymphocyte ratio (NLR); nomograms are unvalidated.

7 · Imaging

Why this matters: a negative scan never clears the groin.

- **Primary**: examination is reliable. Ultrasound (US) sees the tunica but cannot separate T1 from T2. **MRI**, with or without artificial erection (PGE1), helps if equivocal and organ sparing is planned. CT is not useful.
- **cN0 groin**: **no imaging detects micrometastases**, so stage surgically; CT only if obese or prior groin surgery.
- **cN+ or T3–T4**: CT chest, abdomen and pelvis; central necrosis or irregular border predicts poor pathology. **FDG-PET/CT** is best for pelvic and distant disease.

8 · Staging: TNM 8th edition (2017)

Why this matters: T1a versus T1b decides who needs groin staging.

T	Definition
Tis	PeIN
Ta	Non-invasive localised SCC (incl. pure verrucous without destructive invasion)
T1	Glans: lamina propria · Foreskin: dermis/lamina propria/dartos · Shaft: tissue between epidermis and corpora
T1a	**No LVI, no PNI, not high grade**
T1b	**LVI and/or PNI, or high grade (grade III/sarcomatoid)**

T	Definition
T2	**Corpus spongiosum**, with or without urethra
T3	**Corpora cavernosa** (incl. tunica albuginea), with or without urethra
T4	Scrotum, prostate, pubic bone

N	Clinical (cN)	Pathological (pN)
1	Palpable mobile unilateral node	**≤2 unilateral**, no extranodal extension (ENE)
2	Mobile ≥2 unilateral or bilateral	**≥3** unilateral or bilateral
3	Fixed mass or pelvic nodes	**ENE or pelvic nodes**

Stage	I	IIA	IIB	IIIA / IIIB	IV
TNM	T1a N0	T1b, T2 N0	T3 N0	T1–3 N1 / N2	T4, any N3, any M1

- Since the 7th edition: PNI added to T1b; T2/T3 split because cavernosal invasion is worse; **urethral invasion no longer upstages**; pN1 widened to 2 nodes.
- M work-up: liver tests, **calcium**, CT, and bone scan or FDG-PET/CT as indicated.

> **KEY POINT**
> Traps: T1b and T2 share stage IIA; any N3 or T4 is stage IV even without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). The primary is staged by tissue, the groin by surgery.

9 · Management of the primary tumour

Why this matters: cure while keeping as much penis as is safe.

- **Penectomy** is the oncological gold standard (local recurrence **0–8%**); over half of patients are ≤60, so preserve where safe.
- **Organ preservation** suits PeIN, Ta, T1, grade I–II in **compliant** men. **Against**: size **≥4 cm**, grade III, deep cavernosal invasion, or unfit for salvage.
- Local recurrence is higher, but **early recurrence is salvaged without survival cost**, except in pT3 or high grade.
- **Margins**: microscopic spread is **~5 mm** for grade I–II and **~10 mm** for grade III. Only margins **<1 mm** raised recurrence, so a **negative frozen-section margin (>1 mm) is enough**; the 2-cm dogma is abandoned.

Stage / site	Treatment
PeIN, glans	Laser, resurfacing; topical alternative
Ta / PeIN — foreskin, shaft skin	Excision to negative margin; laser; topical (Tis only)
Ta, T1 glans	Excision, resurfacing, glansectomy, radiotherapy (RT; not Ta)
T2 glans, no gross cavernosal involvement	**Total glansectomy** with or without distal corporal transection; partial penectomy; RT
T2 with corporal invasion, T3	**Partial or total penectomy**
T4	Neoadjuvant chemotherapy, then surgery in responders
Recurrence after conservation	Re-excision or penectomy; superficial low-grade re-treated conservatively

- **Glans resurfacing** strips epithelium and subepithelium off the spongiosum (total better than partial) and adds a split-skin graft; erections are kept.
- **Laser**: CO_2 reaches ~0.1 mm (recurrence up to 50%). **Nd:YAG** (1064 nm, up to 6 mm) is preferred, with base biopsies; recurrence <10% for CIS, 10–25% for T1.
- **Mohs** surgery has recurrence ~⅓ and no clear advantage over frozen-section excision.
- † Partial penectomy leaves a stump for standing micturition; total divides the crura with a **perineal urethrostomy**.

> **KEY POINT**
> Margin width depends on grade, not a fixed 2 cm. Low-grade tumours spread millimetres and their recurrences are salvageable, so preservation is safe; in high-grade or cavernosal disease local relapse is lethal and amputation regains its place.

10 · Inguinal nodes *(see Ch 84)*

Why this matters: node status is the strongest prognostic factor.

- P2 P3 Importance and management of Inguinal lymph nodes in CA penis; Prognostic index score in carcinoma penis — *nodes = strongest prognostic factor, Kroon early vs delayed, risk groups, DSNB, FNAC without antibiotic trial, PLND >2 nodes/ENE, neoadjuvant TIP, ILND morbidity*

Not in the paper answer:

- DSNB refinements: pre-operative US with or without FNAC, serial sectioning with cytokeratin, and exploring non-visualising groins.
- Positive pelvic nodes are best treated by neoadjuvant chemotherapy, then surgery.
- Cure is ~**80%** with ≤2 unilateral nodes, no ENE or pelvic nodes; with positive pelvic nodes, 5-year survival after surgery alone is ~**14%**.
- Occult nodes: PeIN/Ta/T1a **0–18%**; T1b+ one-third to over half; corporal invasion ~60%.

> **ALGORITHM**
> 1. **cN0, low risk**: surveillance. If a node appears, do fine-needle aspiration cytology (FNAC); if positive, inguinal lymph node dissection (ILND) and contralateral staging.
> 2. **cN0, high risk (T1b+)**: bilateral dynamic sentinel node biopsy (**DSNB**) or superficial/fascia-sparing/minimally invasive ILND. If positive, completion ILND.
> 3. **cN1–2, mobile, <4 cm**: biopsy, then ILND and contralateral staging. If >2 nodes or ENE, add pelvic lymph node dissection (PLND).
> 4. **Fixed, ≥4 cm or pelvic**: neoadjuvant TIP (cisplatin-eligible) or chemoradiation, then resect responders.
> 5. **Adjuvant therapy** if >2 nodes, bilateral nodes, ENE or pelvic nodes.

- DSNB refinements cut false negatives from ~18% to ~7%. Morbidity: radical > superficial or fascia-sparing > minimally invasive > DSNB.
- No pelvic skip metastases occur; PLND is mainly staging. Reduce morbidity with transverse incisions, saphenous sparing, drain removal by volume, thromboprophylaxis; sartorius transposition *increases* complications.

11 · Radiotherapy

Why this matters: RT keeps the penis.

- **Circumcise first** to expose the lesion and prevent oedema and phimosis. Surgical salvage remains curative.

- **External-beam RT (EBRT)**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks**, with a wax or Perspex block acting as **bolus**; ~55% are recurrence-free at 5 years. Better: >60 Gy, ≤45 days, ≥2 Gy per fraction; worse: T3, >4 cm, high grade.
- **Brachytherapy** (Ir-192) suits **T1–T2 glans tumours <4 cm**: ~60 Gy over ~5 days; ~79% relapse-free and ~85% preservation at 5 years; better erections than EBRT. Meatal stenosis with >60 Gy or >25 cc, ulceration with >65 Gy.
- **Late effects**: **meatal stenosis** (10–45%, 18–24 months) and ulceration; **1 in 5 local recurrences occur after 5 years**. Nodal staging indications are unchanged after RT (biopsy supplies grade and LVI).
- Surface mould: superficial or non-invasive tumours only. Ulceration: conservative care, hyperbaric oxygen, biopsy if suspicious.
- Groins: **no prophylactic RT** for cN0; adjuvant RT for **≥2 nodes or ENE**: ≥50 Gy, **54 Gy for ENE and 57–60 Gy for gross residual**. Chemoradiation (cisplatin if GFR >50 mL/min) treats inoperable nodes.

12 · Systemic therapy

Why this matters: response selects patients for node surgery.

- No randomised trials are completed. **TIP** (paclitaxel, ifosfamide, cisplatin) is most active; TPF is toxic; doublets suit less fit men. **Bleomycin regimens are discouraged**.
- **Neoadjuvant TIP ×4** for bulky N2–N3 or pelvic nodes: ~50% respond, ~⅔ of responders stay disease-free, so **response selects for surgery**.
- Adjuvant chemotherapy has no proven survival gain; consider it for pN3 or pelvic nodes. Cisplatin-ineligible: chemoradiation (mitomycin and capecitabine).
- Palliative platinum is first line (~3× median survival); avoid TIP/TPF with neuropathy or low GFR. Second line: ≤6 months. Anti-EGFR responses are sporadic; checkpoint inhibitors help ~1 in 6 (trials). InPACT is testing sequencing.

13 · Follow-up

Why this matters: groin relapse after observation is usually fatal.

- After organ preservation, use close follow-up and self-examination; RT recurrences can be late.
- Observed groins: 2–3-monthly in years 1–2, 4–6-monthly in year 3, then 6-monthly, then annually. Only ~⅓ of cN0 men with groin relapse survive 5 years. Second primaries occur in 5–15%.

14 · Non-squamous penile tumours (~5%)

Why this matters: each behaves differently.

Tumour	Key features / treatment
BCC	Proximal shaft near the base; almost never metastasises. Treat by excision or Mohs
Melanoma	<1%; glans; spreads by blood and lymph; AJCC cutaneous staging. Circumcision, partial or total penectomy by site. Bilateral modified ILND if Breslow ≥1 mm, ulceration or Clark IV–V (DSNB reasonable); adjuvant nivolumab for stage IIIB–IV

Tumour	Key features / treatment
Sarcoma	Malignant are **proximal**; vascular commonest; mimics Peyronie or priapism; **local recurrence characteristic**. Superficial: wide excision or partial; deep corporal: total. No ILND unless palpable
Extramammary Paget	Penoscrotal eczematoid plaque; CK7+, CEA+, S-100−; ~½ have another cancer, so screen. Wide excision (up to 3 cm) guided by **mapping biopsies**
Lymphoma	Usually systemic: R-CHOP even if isolated
Metastases	From **bladder, prostate, rectum**; **priapism** is the commonest sign; most die within a year (prostate origin longer). Palliation: penectomy for pain or solitary distal disease, dorsal nerve section, suprapubic cystostomy

15 · Landmark trials

Why this matters: they underpin early staging and neoadjuvant TIP.

Trial	Design	Result	Take-home
Neoadjuvant TIP (MD Anderson)	Phase II; TIP ×4, cN2–N3 M0	Response 50%; median OS 17.1 months; durable mainly in responders	Standard before surgery for bulky/pelvic nodes
Netherlands Cancer Institute early vs delayed	Early DSNB vs dissection after surveillance relapse	3-y survival 84% vs 35%; ENE near-universal if delayed	Stage high-risk cN0 groins early

84 Inguinal Node Dissection

1 · Why the groin matters in penile cancer

Why this matters: groin surgery alone can cure node-positive disease.

- Penile squamous cell carcinoma (SCC) has a **locoregional phase**, so the groin is a **curable station**.
- **Nodal status is the strongest prognostic factor**. Extranodal extension (ENE), even in one node, makes it pN3.
- Early versus delayed inguinal lymph node dissection (ILND) for impalpable T2–3 nodes: ENE ≈20% vs ≈95%; 3-year disease-specific survival ≈84% vs ≈35%. So manage the groins within **3 months** of diagnosis.
- An adequate dissection removes ≥**8 nodes per side**. Omitting groin surgery predicts mortality.

2 · Indications and risk groups

Why this matters: primary pathology decides the impalpable groin's fate.

	Low risk	Intermediate	High risk
NCCN	pTis, pTa, pT1a	pT1b G1–2	≥pT2, or G3/G4
EAU-ASCO	pTa, pTis, G1 pT1a	G2 pT1a	≥pT1b
cN0 — NCCN	Surveillance	Modified/radical bilateral ILND or DSNB	Same
cN0 — EAU-ASCO	Surveillance	Surveillance (compliant) vs DSNB	DSNB; ILND if unavailable

NCCN = National Comprehensive Cancer Network; EAU-ASCO = European Association of Urology–American Society of Clinical Oncology; DSNB = dynamic sentinel node biopsy; G = grade. **Trap**: pT1b G2 is intermediate (NCCN) but high (EAU-ASCO).

- For the impalpable high-risk groin, **NCCN leads with ILND** (DSNB if experienced) and **EAU-ASCO leads with DSNB** (ILND if unavailable or preferred). Low risk: surveillance, no staging imaging.
- Select on **primary pathology only** (T, grade, lymphovascular or perineural invasion, sarcomatoid), not HPV or nomograms, with expert pathology review.
- **cN+**: about half of palpable nodes are metastatic. Do fine-needle aspiration cytology (FNAC); if negative, excisional biopsy. If both are negative, survey; if any is positive, do **bilateral radical ILND**. Give no antibiotic trial.
- With unilateral palpable nodes, the other side has micrometastasis in ≈**30%**, so stage it.
- **Bulky or fixed (cN3)**: give cisplatin–taxane chemotherapy first, then surgery. Resect upfront only if chemotherapy-ineligible or vessel invasion is imminent. If skin is involved, do wide excision with a gracilis or rectus myocutaneous flap.

3 · Staging work-up

Why this matters: every test can miss small deposits.

- Indeterminate examination needs cross-sectional imaging. Palpable nodes or high risk need ^{18}FDG-PET/CT or CT chest–abdomen–pelvis. Imaging misses deposits <1 cm, so it has no role in the impalpable groin.
- **FNAC** is specific but insensitive: if positive, ILND; if negative, DSNB or ILND, never discharge.
- **DSNB**: do ultrasound (US), with or without FNAC, first, because a **tumour-replaced node** reroutes tracer to a neo-sentinel node and gives a false negative. Then use ^{99m}Tc nanocolloid and blue dye. Glans lesions may drain straight to deep nodes (another false negative). Use serial sectioning. Positive: ILND †; negative: surveillance.

> **KEY POINT**
> Every groin test fails toward **false negatives**; a negative result never ends the pathway.

4 · Lymphatic anatomy

Why this matters: landmarks keep the dissection complete and safe.

- P2 Importance and management of Inguinal lymph nodes in CA penis. — *superficial/deep systems, bilateral stepwise no-skip drainage, Daseler's five zones (superomedial = sentinel), superficial 4–25 and deep 1–3 nodes, Cloquet*

Not in the paper answer:

- Pelvic spread needs ipsilateral inguinal disease. Basset first described ILND (1912); Daseler described radical ILND and the five zones (1948).
- Crossover to the contralateral pelvis is extremely rare. Cloquet node lies between the femoral vein and lacunar ligament.
- The **femoral triangle** is bounded by the inguinal ligament, **adductor longus** medially and **sartorius** laterally.
- Raise flaps deep to **Camper's fascia**, which holds the capillary network. Too thin causes necrosis; too thick leaves nodes. Incise along Langer's lines.
- In the fossa ovalis the order, lateral to medial, is **nerve, artery, vein**, so **never dissect lateral to the femoral artery**.

5 · Templates

Why this matters: smaller templates cut morbidity in staging.

Template	Extent	Features
Radical	Whole femoral triangle, superficial and deep	Fascia lata opened; saphenous ligated; sartorius transposed
Modified (Catalona)	Excludes lateral to femoral artery and caudal to fossa ovalis	Shorter incision, saphenous preserved, thicker flaps, no transposition
Superficial	Above fascia lata only	Fascia lata preserved

Modified ILND nodes go to frozen section; if **positive, convert to radical** †. Negative superficial nodes allow the deep dissection to be omitted.

6 · Open radical ILND

Why this matters: the steps protect vessels and flaps.

ALGORITHM

1. Frog-leg position; incision ≈2 cm below and parallel to the inguinal ligament.
2. Raise flaps 3 cm above the ligament to the apex; divide the saphenous at the apex (spare if disease minimal).
3. Open fascia lata over sartorius and adductor longus; dissect cephalad on the femoral vessels.
4. Divide the saphenous at the junction; clear deep nodes up to the femoral canal.
5. **Sartorius transposition**; close the femoral canal (inguinal to Cooper's ligament).
6. Suction drains; tack the flaps.

- A vessel sealer is preferred to monopolar; fibrin sealant gives no benefit.

7 · Pelvic lymph node dissection (PLND)

Why this matters: pelvic nodes select adjuvant therapy.

- If the inguinal frozen section is positive, do PLND at the same sitting; otherwise stage it.
- † EAU: with ≥2 positive inguinal nodes or ENE (pN2–3), do ipsilateral PLND, open or minimally invasive.

8 · Video-endoscopic ILND (VEIL)

Why this matters: VEIL avoids the problem groin wound.

- P4 Videoendoscopic inguinal lymphadenectomy — *same template beneath Scarpa's, 15 mm Hg, frozen section gates deep dissection; similar yield, fewer wound complications; non-bulky nodes only*
- Ports sit ≈25 cm below the ligament; check haemostasis at 5 mm Hg. Severe oedema is rarer; the commonest robotic complication is lymphocele.

9 · Complications and post-op care

Why this matters: fear of these must not cause omission.

Complication	Cause	Prevention / management
Wound infection, breakdown (commonest)	Skin flora, dead space	Asepsis, tack flaps, incisional negative pressure
Flap necrosis	Breach of Camper's capillary network	Correct plane, Langer-line incision, modified template, VEIL; debride, then flap or negative pressure †
Lymphocele / seroma	Unsealed lymphatics	Sealing devices; drain until **<30 mL/day**; drain symptomatic or infected collections †
Lymphoedema	Lymphatic interruption	Saphenous sparing, modified template, compression hose ≥1 month
Others	Phlebitis, neurapraxia, pulmonary embolism (PE), femoral bleeding	Thromboprophylaxis; stay medial to artery

Mobilise early. NCCN advises antibiotic prophylaxis (oral cephalosporin for days to weeks); the authors do not routinely. The root causes are disrupted **skin blood supply** and **lymphatics**; high-volume surgeons reduce both.

10 · Follow-up

Why this matters: most groin relapses come early.

Median recurrence is ≈1 year, and most occur within 2–3 years. EAU-ASCO: examine 3-monthly for 2 years, then 6-monthly to 5 years; image only if node-positive. NCCN images everyone (CT abdomen–pelvis and chest X-ray), even pN0. Follow more closely after DSNB; self-examination after 5 years.

83 Tumours of the Urethra

1 · Benign urethral tumours

Why this matters: these mimic cancer but need only local treatment.

Lesion	Who / where	Clue	Treatment
Leiomyoma	Reproductive-age women; posterior wall	Firm suburothelial mass; grows in pregnancy	Excision; may recur
Haemangioma	Men, any age	Bluish sessile lesion, painless bleeding; posterior lesions cause haematospermia; Klippel–Trénaunay–Weber	Fulguration/laser; open excision if extensive
Fibroepithelial polyp	Boys; bulbar/posterior	Polyp on a stalk; filling defect	Transurethral resection

2 · Anatomy, histology and lymphatics

Why this matters: tumour site predicts its type and its nodal basin.

Feature	Male (~20 cm)	Female (~4 cm)
Segments	Anterior: fossa navicularis, pendulous, bulbar. Posterior: membranous, prostatic	Anterior = distal ⅓; posterior = proximal ⅔
Epithelium	Urothelium in the prostatic urethra, columnar in the anterior urethra, squamous in the fossa navicularis	Proximal ⅓ urothelium; distal ⅔ stratified squamous
Glands	Cowper's, Littré	Skene's (prostate homologue)
Histology	Urothelial carcinoma (UC) 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%. **Anterior: up to 80% SCC; posterior: mostly UC**; adenocarcinoma relatively more in bulbar	UC 28–45%, SCC 19–29%, **adenocarcinoma 28–38%** (near-equal)
Lymphatics (both sexes)	Anterior drains to **inguinal** nodes (bulbar also external iliac); posterior drains to **pelvic** nodes	

KEY POINT

Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women it is also the **continence boundary**: the distal third can be excised without incontinence.

- Grading: UC is low or high grade; SCC and adenocarcinoma are G1–G3.

3 · Staging: AJCC 8th edition

Why this matters: the prostatic urethra has its own T categories.

T	Penile urethra and female urethra	Prostatic urethra
Ta / Tis	Non-invasive papillary / carcinoma in situ (CIS)	Tis also includes prostatic duct involvement **without stromal invasion**
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma

T	Penile urethra and female urethra	Prostatic urethra
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (bladder)	Bladder, rectum
N	N1 single, N2 multiple regional nodes (inguinal, pelvic, presacral): **number, not size**	

- **Evaluation**: examination under anaesthesia (EUA) with cystourethroscopy, bimanual palpation and biopsy; cytology is insensitive. Use **MRI** for local extent and CT chest if invasive.
- **Palpable inguinal nodes (20–30%) are almost always metastatic**, so, unlike penile cancer, no antibiotic trial is given.

4 · Male primary urethral cancer (PUC)

Why this matters: distal tumours are curable by surgery, while bulbar tumours need multimodal therapy.

- Patients are >60 years. **Urethral stricture** is the commonest risk factor (up to 50%; latency 20–30 years). Others are lichen sclerosus, sexually transmitted infection (STI), pelvic radiotherapy (RT) and human papillomavirus (HPV) 16.
- † It may present as a periurethral abscess or fistula. † Biopsy a stricture that bleeds or recurs rapidly.
- **Prognosis** depends on stage and location: distal tumours do 2–3-fold better, and bulbar survival is as low as 20–30%.

Pendulous / fossa navicularis (distal)

- Surgery alone controls low-stage disease. The ladder runs from transurethral resection, to **distal urethrectomy with or without partial penectomy**, to total urethrectomy with perineal urethrostomy, with or without total penectomy.
- **Penile preservation** is endorsed by the European Association of Urology (EAU) and National Comprehensive Cancer Network (NCCN): topical 5-fluorouracil (5-FU) for distal CIS, distal urethrectomy with two-stage urethroplasty, glansectomy with graft, or anterior urethrectomy with perineal urethrostomy. A positive margin needs re-excision or adjuvant RT.
- A **margin of 5 mm** is now standard (1 cm traditionally). Invasion into the proximal penile urethra needs total penectomy. Distal CIS can be upstaged.
- **Primary RT** is only for early anterior lesions in men refusing surgery; it causes stricture, ulceration and oedema. It is ineffective for bulbar PUC.

Bulbar (proximal anterior)

- The few low-stage tumours are treated by transurethral resection, or segmental excision and anastomosis.
- **Advanced** disease needs radical cystoprostatectomy, pelvic lymphadenectomy, total urethrectomy and penectomy, with or without the inferior pubic rami and urogenital diaphragm. Surgery alone gives ~¼ 5-year survival, so use **multimodal therapy**.

Inguinal nodes (both sexes)

- There is **no proven benefit of prophylactic inguinal dissection in cN0**. If the primary was irradiated and the groin is cN0, consider prophylactic groin RT.
- For cN1, give neoadjuvant chemotherapy or chemoradiotherapy, then lymphadenectomy.

5 · Male posterior (prostatic) urethral cancer

Why this matters: only stromal invasion needs cystectomy.

- It is usually **secondary to bladder UC**: 20–40% at radical cystectomy (RC), and up to 39% of high-risk non-muscle-invasive bladder cancer (NMIBC) recurs in the prostate after intravesical therapy. So always evaluate the bladder and upper tracts.
- Diagnose by transurethral resection of the prostate (TURP) biopsy; it is often CIS. Bacillus Calmette–Guérin (BCG) for high-grade superficial disease gives ≥70% response; extensive TURP with BCG gives ~half complete response.

Involvement	Treatment	5-year survival
Mucosal	TURP + BCG	67–74%
Ductal / acinar (still Tis)	TURP + BCG, or RC with or without urethrectomy	67–74%
Stromal invasion	**RC + urethrectomy + pelvic lymphadenectomy**, with or without neoadjuvant chemotherapy	~36%; follows bladder stage

KEY POINT

The only question is "stroma breached?" TURP before BCG debulks, improves contact and stages. Ductal disease is not T2, so do not jump to cystectomy.

6 · Female urethral cancer

Why this matters: location decides whether continence can be kept.

- It is ~⅓ of PUC, in women >60 years. Risks are chronic irritation, infection, leukoplakia, caruncle, HPV and **urethral diverticulum (6–10% harbour malignancy**, mostly adenocarcinoma).
- Distal tumours are SCC; proximal ones are UC or adenocarcinoma. Adenocarcinoma is **mucinous** (commoner; CEA) or **clear cell** (in diverticula; may secrete PSA), probably of Skene's origin.
- **Location decides continence-sparing excision**; ⅓ are distal. Stage with MRI pelvis.
- 30–50% are locally advanced and ~20% have palpable inguinal nodes. **Stage dominates prognosis**: distal is better than proximal, which is better than pan-urethral. Melanoma is uniformly poor.

Treatment

Situation	Management
Small non-invasive	Endoscopic resection or laser, with or without BCG
Low-stage distal	**Distal urethrectomy** (transvaginal) **or RT** (brachytherapy with or without external beam, ~60 Gy): equal, ≥70% cured. Stress incontinence (>40%), meatal stenosis; ~1 in 5 recur locally. RT complications 20–50% (stricture, necrosis, fistula)
Proximal / pan-urethral	**Anterior exenteration**: RC, total urethrectomy, extended pelvic lymphadenectomy and anterior vaginal wall, with or without hysterectomy, vulvectomy, pubic symphysis/rami. Alone, most die and ~⅓ recur in the pelvis, so use **multimodal** therapy
Palpable groin nodes	Chemoradiotherapy, chemotherapy or lymphadenectomy; often curable. Pelvic nodes are poor

- RT local control depends on the **length of urethra involved**, not histology or size; external beam plus brachytherapy beats external beam alone.

7 · Multimodal therapy (both sexes)

Why this matters: surgery or RT alone fails in advanced and proximal disease.

- For advanced or proximal disease, give **cisplatin-based neoadjuvant chemotherapy** (EAU, NCCN) or **chemoradiotherapy**, then consolidative surgery. Response predicts survival.
- **Chemoradiotherapy** is 45–55 Gy to the genitalia, perineum and groins with 5-FU and mitomycin C. It is preferred for advanced SCC; every responder develops a stricture and ~⅓ recur locally. It is not for low-stage distal disease.

Histology	Regimen
UC	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin); gemcitabine–cisplatin
UC or SCC	Paclitaxel, ifosfamide, cisplatin (TIP); cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gemcitabine, 5-FU, leucovorin, cisplatin

Stage (NCCN, anterior male / female)	First line
Tis, Ta, T1	Repeat transurethral resection, with or without intraurethral BCG or chemotherapy
Male T2 pendulous	Partial urethrectomy with or without partial penectomy; if margin positive, chemoradiotherapy, surgery or RT
Male T2 bulbar	Urethrectomy with or without RC; if ≥pT3/N1, adjuvant chemotherapy or chemoradiotherapy
Female T2	RC + urethrectomy, or distal urethrectomy by location, or chemoradiotherapy
≥T3 N0	Chemoradiotherapy with or without surgery; or neoadjuvant chemotherapy, then surgery or RT
N1	Chemoradiotherapy (preferred for SCC) or systemic therapy, with or without consolidative surgery

- Recurrence after resection of invasive disease is ~50%, so survey the urethra, groins and body. PD-L1-positive tumours may respond to checkpoint inhibitors.

8 · Urethral recurrence (UR) after radical cystectomy

Why this matters: urethral risk decides whether a neobladder is safe.

Male

- Risk is **1–15%** (~5%); most occur within 2 years, but ~11% after 5 years.
- **Risk factors**: **prostatic urethral involvement** (15% vs 5%), **positive urethral margin**, cutaneous diversion, CIS, multifocality and prior NMIBC. An orthotopic neobladder (ONB) is protective (~5% vs 9%).
- Routine prophylactic urethrectomy is abandoned, because 64–100% of specimens are negative.
- **Before ONB**, confirm a negative intraoperative frozen section of the urethral margin; this is better than pre-op prostatic biopsy.

- **Urethrectomy** is absolute for a positive margin or gross urethral tumour, and relative for diffuse CIS, prostatic urethral involvement, multifocal disease, or bladder neck or upper tract tumours. Delayed urethrectomy is as good as simultaneous.

Female

- Synchronous urethral involvement is 8–13%. The main risk is **bladder neck involvement**, but only 30–50% of these have urethral tumour, so it is not an absolute contraindication.
- ONB is allowed if the bladder neck, urethra and anterior vaginal wall are uninvolved and frozen section is negative. UR after ONB is <5%; most come from skip lesions despite negative margins, so keep up surveillance.

Surveillance and treatment

- Urethral wash cytology (barbotage after cutaneous diversion) for 2 years has no proven survival benefit. If positive, do urethroscopy and biopsy.
- **Total urethrectomy is treatment of choice**, including the meatus; after ONB, convert to cutaneous diversion. Non-invasive UR can be resected endoscopically (high second-UR risk); **intraurethral BCG for CIS** (not papillary or invasive).
- Invasive UR has median survival ~17 months.

85 Benign Disorders of the Penis and Urethra

1 · Reconstructive surgical technique

Why this matters: reconstruction depends on the wound bed's blood supply.

- **Cautery**: **bipolar on the urethra** (minimal spongiosal damage); monopolar on subcutaneous tissue and muscle.
- A **bougie à boule** calibrates. **Goodwin sounds** dilate and are safer than van Buren, as they pass over a guidewire.
- **Sutures** are always **absorbable** (permanent ones form a stone nidus); monofilament polydioxanone (PDS) is preferred. Support is needed for **30–60 days**; chromic lasts 14 days, so avoid it. Use **5-0** for most urethral work and **6-0** for a single-layer mucosal anastomosis (non-transecting excision and primary anastomosis, EPA), on an RB-1 taper needle.
- Limit high lithotomy (neuropathy, compartment syndrome); low lithotomy for graft harvest.

2 · Tissue transfer: grafts and flaps

Why this matters: graft or flap choice is the core of urethroplasty.

- P1 Enlist various grafts used in urology and role of BMG in urology — *graft vs flap; take by imbibition then inosculation (~48 h each), immobilise 5–10 days; STSG vs FTSG; other grafts; BMG properties, buccal = lingual*

Not in the paper answer:

- The graft is below core temperature during imbibition and reaches core after inosculation. Poor bed vascularity impairs take.
- Skin in contact with urine must be **hairless** (laser or electrolysis before harvest).
- **Extragenital full-thickness skin graft (FTSG)** fails in the urethra (greater mass makes take ischaemic). The exception is the **posterior auricular (Wolfe) graft**: thin, on perforators, it behaves like a split-thickness skin graft (STSG).
- Oral mucosa graft (OMG) has a **panlaminar plexus** and sacculates less than skin. Buccal mucosa graft (BMG) gives more early swelling and numbness, lingual gives speech difficulty, labial the worst morbidity. Buccal onlay failures occur in the **first 12 months**.

> **KEY POINT**
> Graft trade-off: a dense fine plexus and low mass favour take, while carried dermis or lamina propria gives durability and resists contraction. This explains STSG vs FTSG, the failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal.

Flaps (four independent axes)

- **Composition**: fasciocutaneous (distal penile circular), fascial (dartos), musculocutaneous (rectus), muscle (gracilis) or visceral (omentum).
- **Blood supply**: axial (named vessel: gracilis, penile circular flap) or random (U-flap, **Z-plasty**). Axial flaps may be peninsula, island (skin divided, pedicle intact) or free.
- **Movement**: advancement (Y-V), rotation (Martius) or transposition (Z-plasty). **Location**: local, distant pedicled or free (radial forearm phalloplasty).

3 · Surgical anatomy

Why this matters: fascial layers decide flap planes and how blood or infection spreads.

- Layers, deep to superficial: **tunica albuginea, Buck's fascia, dartos, skin**. Buck's encloses the deep dorsal vein, dorsal arteries and nerves, and splits around the spongiosum. Dartos is fat-free, carries the skin vessels and fuses with Colles fascia.
- The urethra is **eccentric (dorsal) in the bulb** and central in the shaft; the membranous urethra lies within the striated external sphincter.
- **Skin supply**: the superficial external pudendal artery feeds a subdermal plexus (basis of the McAninch circular flap); the posterior scrotal artery feeds the ventral longitudinal flap.
- **Deep supply**: the internal pudendal artery continues as the common penile artery, giving the **bulbourethral, dorsal (circumflex branches, glans) and cavernosal** arteries.

Situation	Spread
Buck's intact (penile fracture)	Confined to the shaft, giving an **eggplant deformity**
Buck's ruptured (straddle)	Within Colles–Scarpa: scrotum, perineum, abdominal wall; limited on the thigh by fascia lata
Fournier gangrene	From perineum and scrotum up the abdominal wall via Scarpa

> **KEY POINT**
> Two envelopes, two vascular worlds: dartos carries the skin supply (so skin flaps are raised on dartos); Buck's encloses the erectile bodies fed by the common penile artery.

4 · Lichen sclerosus (LS)

Why this matters: LS starts at the meatus and can become panurethral.

- P1 Lichen sclerosus (BXO, balanitis xerotica obliterans) — *meatus sentinel, Littre-gland stricture mechanism, spares proximal urethra, biopsy features, RUG first, clobetasol, oral mucosa never skin, perineal urethrostomy*

Not in the paper answer:

- Malignant transformation reaches up to **8%**, mostly squamous cell carcinoma (SCC).
- **Steroid**: clobetasol propionate or betamethasone 0.05% once or twice daily for 1–2 months, then on alternate days for 3–4 months.
- Meatal stenosis: extended meatotomy with an inverted-V relaxing incision (prevents puckering) or BMG onlay; restenosis is commoner than in non-LS. Early meatal or fossa repair may avert a panurethral stricture.
- Panurethral: perineal urethrostomy or staged repair (stricture first, skin later).

> **KEY POINT**
> LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, retrograde urethrography (RUG) before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy.

5 · Other urethral lesions

Why this matters: these rare lesions mimic stricture or tumour.

Lesion	Key features	Management
Urethral haemangioma	Bloody discharge; meatus common; often extends beyond view, so image with **MRI**; benign	Observe; small: laser or triamcinolone; large: excision, staged repair (no tubed grafts)

Lesion	Key features	Management
Reactive arthritis (Reiter)	Arthritis, conjunctivitis, urethritis after Chlamydia/Shigella/Salmonella; HLA-B27 in ~2/3; **circinate balanitis**	Urethritis is self-limiting
Amyloidosis	Urethral mass, obstruction, haematuria; needs biopsy	Observe; dilate or urethroplasty if obstructive
Urethrocutaneous fistula	Mostly iatrogenic (hypospadias, phalloplasty); also piercing, infection behind a stricture	Treat the **underlying process** as well as the tract
Male urethral diverticulum	Congenital (all layers) or acquired (no smooth muscle; behind stricture, catheter, trauma); dribbling, urinary tract infection (UTI), stone	Cystourethroscopy and RUG; repair and treat the cause
Enlarged utricle	Müllerian remnant, with proximal hypospadias	A catheter may enter the utricle; UTI, post-void dribble

6 · Prepuce and meatus

Why this matters: paraphimosis is an emergency.

- **Paraphimosis (emergency)**: a trapped foreskin causes oedema, which can lead to glans necrosis; it classically follows catheterisation. Reduce by steady compression: thumbs push the glans, fingers pull the foreskin (granulated sugar helps). If irreducible, do an emergency **dorsal slit** or circumcision; elective circumcision for recurrence.
- **Phimosis** is physiological in infants (resolves at 1–8 years); in adults it leads to balanitis and cancer. Treat with steroid, dorsal slit or circumcision.
- **Balanitis**: **Candida** is commonest; others are sexually transmitted infection (STI), chemical and Zoon. Treat with hygiene and antifungal, and **check blood glucose**. Recurrent balanitis raises cancer risk.
- P2 Management of Urethral meatal stenosis. — *adult causes, RUG first, ventral meatotomy with 3 sutures, steroid self-dilation, urethroplasty if fossa involved*

Adult circumcision (sleeve)

> **ALGORITHM**
>
> 1. Penile block. Foreskin retracted: mark the cuff incision across the frenular base; incise to Buck's.
> 2. Foreskin advanced: mark the coronal incision, with or without frenuloplasty.
> 3. Excise the sleeve using **bipolar**, because adult vessels do not seal by pressure.
> 4. Quadrant dartos sutures, then skin closure.

- **Human immunodeficiency virus (HIV)**: the foreskin has many CD4+ cells and microtears. Circumcision lowers female-to-male HIV by ~**60%**, and also lowers herpes simplex (HSV) and human papillomavirus (HPV). In men who have sex with men (MSM), benefit is mainly for insertive partners.

7 · Penile curvature

Why this matters: the type decides what is corrected.

- P1 Congenital Penile curvature.. — *chordee without hypospadias vs congenital curvature (young man, large erection); never divide a urethra reaching the glans tip*

Not in the paper answer:

- Congenital curvature is a smooth whole-shaft curve; chordee without hypospadias shows a hooded prepuce and ventral inelasticity.

Devine–Horton type	Abnormal tissue
I	Spongiosum deficient (chordee without hypospadias, I–III)
II	Dysgenetic Buck's and dartos; urethra/spongiosum normal
III	Dartos only
IV	All normal; asymmetric tunica = **congenital curvature**
V	Congenital short urethra; its existence is doubted

ALGORITHM

1. Excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum.
2. Artificial erection; ventral midline septotomy.
3. If still curved, do dorsal tunical ellipse excision or plication under the elevated neurovascular bundle (dorsal arteries may be the dominant glans supply).

- **Acquired non-Peyronie curvature** *(see Ch 76)*: deep urethrotomy into the tunica, neglected fracture, **subclinical fracture** (buckling leaves a lateral scar and indentation) or septal tear (mimics Peyronie disease, PD). Erections are normal, with no shortening. Treat by **scar excision and graft**, not contralateral plication.

8 · Penile transplantation

Why this matters: an option after penile loss.

A composite allotransplant gives natural appearance, erection and voiding, but needs lifelong immunosuppression and raises ethical issues. The alternative is phalloplasty.

9 · Adult acquired buried penis (AABP)

Why this matters: it is self-perpetuating and usually needs surgery.

- Risks: **obesity**, aggressive circumcision, scrotal lymphoedema, LS.
- A **vicious cycle**: the escutcheon tethers the penis, trapped urine causes inflammation, LS and scarring, and these trap more. So **weight loss alone rarely cures**.
- The **Pariser** classification grades the repair needed: I skin flap, II graft, III scrotoplasty, IV escutcheonectomy, V panniculectomy.
- LS cicatrix alone: try topical steroid. CT if hernia suspected. **Evaluate the urethra**: a severe stricture needs urethroplasty **before** grafting. Optimise diabetes.

ALGORITHM

1. Slit the phimotic ring and place glans traction.
2. Circumcising incision; excise non-viable skin.
3. Do escutcheonectomy (harvest the graft from it; keep 2–3 cm of skin at the base).
4. Tack Scarpa to the rectus fascia or pubis; place drains.
5. Cover with STSG (0.012–0.020 inch), fibrin glue and a bolster for 5–7 days.

- Dressing off ~day 5; drains out at <30 mL/day.
- **Scrotal lymphoedema**: congenital (Milroy in infancy, Meige at puberty, tarda in adults), infective (**Wuchereria bancrofti**, lymphogranuloma venereum), hidradenitis, malignancy,

node dissection or radiation. Resect **all** involved tissue; cover with spared perineal or lateral scrotal skin.

- **Foreign body injection** (silicone, paraffin) causes fibrosis, necrosis, abscess and Fournier gangrene. Excise all and graft (STSG); screen for **body dysmorphic disorder**.

10 · Landmark trials

Why this matters: circumcision prevents HIV.

Trial	Design	Result	Take-home
African circumcision trials (South Africa, Kenya, Uganda)	Three randomised controlled trials (RCTs), adult male circumcision vs none	~60% reduction in female-to-male HIV acquisition	Circumcision is an HIV-prevention strategy

86 Urethral Stricture Disease

1 · Definitions, anatomy, aetiology

Name, blood supply and cause decide the operation.

- P2 P3 Urethral stricture disease: etiology, presentation and management (including recent advances); Urethral stricture: classification, DVIU and surgery after failed VIU — *anterior-only definition, spongiofibrosis, area ∝ diameter², Jordan spectrum, dorsal eccentric bulb, dual supply, aetiology table (idiopathic ≈ iatrogenic), LS via Littre glands*

Not in the paper answer:

- Straddle strictures often present late, at obstruction; prostate cancer treatment causes posterior stenosis; Ureaplasma has no link.

Site	Correct term
Anterior urethra	Stricture (epithelial scar with or without spongiofibrosis)
Membranous urethra after pelvic fracture	Distraction defect; pelvic fracture urethral injury (PFUI)
Posterior urethra after trauma/radical prostatectomy	Posterior urethral stenosis
Prostatic urethra, bladder neck	Contracture or stenosis

- The common penile artery supplies the corpus spongiosum. The bulbar urethra lies dorsally, so a deep 12 o'clock urethrotomy in the bulb is dangerous.
- Congenital stricture occurs in infants before they walk. In lichen sclerosus (LS), microabscesses form in the glands of Littre; antibiotics relieve symptoms, not the stricture.

> **KEY POINT**
> Retrograde glans flow allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation causes ischaemia, which links vessel-sparing excision and PFUI repair failure.

> **KEY POINT**
> Aetiology predicts biology: iatrogenic and straddle strictures are focal ischaemic or crush scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it.

2 · Presentation and evaluation

The plan needs the scar's site, length and depth.

- **Acute retention**: if a catheter fails, do a retrograde urethrogram (RUG) and place a **suprapubic cystostomy**, not blind dilation. Filiforms are condemned; pass a guidewire under flexible endoscopic vision.
- Four determinants guide treatment: **location, length, depth, density**.
- The treating surgeon does a dynamic RUG and voiding cystourethrogram (VCUG) (McCallum criteria). Sonourethrography may underestimate spongiofibrosis. Do not dilate at first flexible urethroscopy. Magnetic resonance imaging (MRI) is for tumours and PFUI only.
- **Hydraulic trap**: voiding pressure holds the diseased urethra above the stricture open; a segment that cones down is scarred.

- **Urethral rest** with a suprapubic tube for **6–8 weeks** reveals the true length. At surgery, calibrate the whole urethra (bougie à boule).

3 · Decision making

The first operation has the best chance of cure.

- First agree on **palliation or cure**; periodic dilation is acceptable if agreed. The **reconstructive ladder is archaic**: go to the procedure most likely to cure.

ALGORITHM

1. If bulbar, <1.5 cm, without dense spongiofibrosis and untreated, then dilate or do one internal urethrotomy; if it recurs, then urethroplasty.
2. If bulbar 1–2 cm (up to 3–5 cm proximally), then excision and primary anastomosis (EPA).
3. If longer bulbar, then oral mucosa graft (OMG) onlay, with or without augmented anastomosis.
4. If penile, then onlay (graft or flap); if hostile bed, LS or failed hypospadias, then staged OMG.
5. If panurethral, options exhausted or elderly, then perineal urethrostomy.

4 · Dilation and internal urethrotomy (DVIU)

Endoscopy cures few; repeating it harms later surgery.

- P3 Urethral stricture: classification, DVIU and surgery after failed VIU — *secondary-intention race, 10–2 o'clock thin spongiosum, complications, second urethrotomy futile, self-dilation, mitomycin C, laser = cold knife, catheter 3–7 days*

Not in the paper answer:

- Bleeding during dilation means tearing; the safest method is a **balloon** over an endoscopically placed guidewire.
- Self-catheterisation is valid palliation. Deep cuts can enter the intracrural space and cause erectile dysfunction (ED).
- Dilation cures only an epithelial stricture without spongiofibrosis. **Optilume**: paclitaxel balloon (see trial).
- Direct vision internal urethrotomy (DVIU) does worse when luminal narrowing exceeds **74%**. It fails in non-bulbar strictures; repetition worsens later urethroplasty. Use saline (extravasation).

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis
<1.5 cm	Longer
No dense deep spongiofibrosis	Dense (e.g. straddle)
First procedure	Repeat

- Self-catheterisation tapers over **3–6 months**; the stricture recurs when it stops. A 6-week catheter is no better than 3–7 days. UroLume is withdrawn.

KEY POINT

No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; each repetition spends the patient's reconstructive capital.

5 · Excision and primary anastomosis (EPA)

The most durable repair for short bulbar strictures.

- EPA is the **gold standard**; recurrence is rare, mostly within 6 months. Rules: **excise all fibrosis, spatulate widely, keep it tension-free**.
- Length comes from spongiosal mobilisation, freeing Buck's fascia, the intracrural space, and detaching bulbospongiosus from the perineal body (about 1–1.5 cm). **The more proximal the stricture, the longer the excisable segment.**
- Use absorbable 3-0/4-0 sutures for anastomosis, 4-0 to 6-0 for grafts and flaps, and bipolar cautery.

	Transecting	Vessel-sparing transecting	Non-transecting
Bulbar arteries	Divided	Preserved	Preserved
Steps	Excise, spatulate, anastomose	Divide at membranous–bulbar junction, dorsal spatulation	Dorsal stricturotomy, excise scar from within, close transversely
Use	Most	Proximal bulbar	Proximal

Vessel-sparing protects men with poor retrograde supply (hypospadias, trauma, prior surgery); less ED and less artificial sphincter cuff erosion are theoretical and unproven.

6 · Substitution urethroplasty

Long strictures need new tissue; the bed decides success.

- **Buccal mucosa graft (BMG)** is the workhorse (one-stage onlay and staged). Full-thickness skin graft (FTSG) is not used in LS; meshed split-thickness skin graft (STSG) is staged only; Wolfe graft is for staged hypospadias. Grafts do best in the **bulb**, which has a thick vascular bed.
- Graft position: ventral only with spongioplasty; lateral (thin spongiosum, less bleeding); **dorsal (Barbagli)**, spread-fixed on the corpora.
- **Augmented anastomosis** often beats pure onlay. **Kulkarni** is a one-stage panurethral dorsal onlay via the perineum with **penile invagination**.
- **Staged**: stage 1 lays it open and quilts the graft to dartos (not tunica, which cannot be tubed). Stage 2 is Thiersch tubularisation to about 28 Fr after at least 6 months (12 after STSG). Used for LS and failed hypospadias.

> **KEY POINT**
> A graft lives on its bed: it succeeds where the bed is thick and vascular (bulb, dorsal corpora), and is staged when the bed, or skin disease, cannot be trusted in one step.

- **Flaps** use hairless genital skin on dartos; in **Quartey** flaps the skin is a passenger on fascia. Designs: ventral longitudinal (**Orandi**), dorsal transverse, hockey stick (7–9 cm), **circular island** up to **15 cm**; scrotal island is not first choice.
- **Fossa navicularis**: dartos transverse (Jordan) or longitudinal (De Sy) island.
- Grafts equal flaps; flaps suit irradiated or multiply operated urethras. **Onlay beats tube**, and a tubed flap beats a tubed graft. Very long: flap distally, graft proximally.
- **LS**: skin repairs fail, so use **oral mucosa only**, often staged.

- **BMG perineal urethrostomy**: via an inverted-U incision, transect spongiosum and open dorsally to healthy mucosa. Suture ventral urethra to skin (30 Fr); a BMG quilted to the corpora completes the dorsal wall.

7 · Meatal stenosis

Simple in children; in adults it signals deeper disease.

- **Child**: post-circumcision meatitis needs ventral meatotomy and parental dilation 3×/day for 7–10 days; early meatitis responds to 0.05% clobetasol.
- **Adult**: the fossa is usually involved, so evaluate as a stricture. Do a ventral meatotomy with three sutures, or dorsal Y-V if needed.

8 · Pelvic fracture urethral injury (PFUI)

An obliterative gap, repaired by anastomosis, not endoscopy.

- P3 Describe early and late management of PFUD injuries — *bulbomembranous site, primary realignment debate, cystogram + RUG, bladder neck not predictive, MRI, delayed perineal repair after 3 months, progressive steps, voiding study 21–28 days*

Not in the paper answer:

- A normal anterior urethra on RUG predicts a successful anastomosis. Endoscopic **core-through is not curative**.
- Failure is mostly **ischaemic stenosis** of the mobilised bulb, not anastomotic; internal pudendal status decides the order of surgery (table below).
- PFUI occurs in about **10%** of pelvic fractures; "apple off its stem" is wrong. Primary realignment is linked to delayed repair and more endoscopic failures.
- Repair at **3–6 months**; a transpubic approach is unnecessary.

ALGORITHM

1. On-table endoscopy (exclude bladder stones); exaggerated lithotomy; perineal incision.
2. Divide bulbospongiosus and mobilise the bulb; pass a Haygrove sound via the suprapubic tract; excise fibrosis.
3. Spatulate to **≥30 Fr**; place 8–10 tension-free 4-0 polydioxanone (PDS) sutures over a silicone stent.
4. If tension remains, then add steps in order: mobilisation (to penoscrotal junction), intracrural separation, infrapubectomy, corporal rerouting.

- Success is in the high **90s %**. Give an anticholinergic after surgery.

Internal pudendal status	Potency	Action
Intact ≥1 side	Often potent	Reconstruct
Reconstituted only	Never potent	Reconstruct; revascularise for ED later
Bilateral occlusion	—	**Revascularise first** (ischaemic stenosis risk)

Screen at-risk men (dorsal nerve injury, failed repair, lateral-impact fracture) with **duplex ultrasound, then angiography if abnormal**.

9 · Vesicourethral stenosis and fistula

These follow prostate cancer treatment.

- Incise deep at 3 and 9 o'clock; if recalcitrant, radial incision and mitomycin C. Obliteration needs robotic redo or perineal repair (high incontinence; later artificial urinary sphincter, AUS). Suprapubic tube or diversion are always options.
- **Rectourethral fistula**: simple (<2 cm, non-irradiated) is repaired perineally or by York-Mason. Complex (irradiated) needs faecal diversion and transperineal **gracilis interposition, with or without BMG**; after radiation, diversion is often the endpoint.

10 · Complications and follow-up

Recurrence timing differs by repair.

- Recurrence is early after EPA, and within 12 months after grafts, declining with time. ED is about the circumcision rate, higher with long repairs and straddle, and often improves. Chordee and positioning neuropathy occur.
- Follow up with symptoms, †uroflow and post-void residual (PVR), and urethroscopy if suspicious.

11 · Landmark trials

The trial to quote on drug-coated balloons.

Trial	Design	Result	Take-home
ROBUST III	Phase III randomised controlled trial (RCT), Optilume vs endoscopic control after multiple failed endoscopic treatments	1-year stricture-free 74.6% vs 26.8%	Drug-coated balloon beats repeat endoscopy; role as first-line undefined

87 Surgery of the Scrotum and Seminal Vesicles

1 · Surgical anatomy of the scrotum

Lax skin and a triple blood supply shape scrotal surgery.

- Lax, rugated skin gives **poor tamponade**, so haematomas keep enlarging; haemostasis must be meticulous.
- Layers from testis outwards: albuginea, tunica vaginalis, internal spermatic fascia, cremaster, external spermatic fascia, dartos, skin. Dartos keeps necrotising fasciitis off the testes.
- **Transdermal absorption** is about **40×** that of other skin, and the skin is rich in 5α-reductase.

Territory	Artery	Parent
Scrotum, anterior	Superficial and deep external pudendal	Femoral
Scrotum, posterior	Posterior scrotal	Perineal (internal pudendal)
Testis, primary	Testicular	Aorta
Testis, secondary	Deferential	Superior vesical, from internal iliac
Testis, secondary	Cremasteric	Inferior epigastric, from external iliac

KEY POINT

Three trunks (aorta, internal iliac, external iliac) anastomose in the cord (about 10 cm above the testis) and at the cauda. So the testicular artery can be ligated in Fowler–Stephens orchidopexy, and the superior epididymal artery sacrificed in vasectomy reversal.

- Nerves: the genital branch of the genitofemoral (injury abolishes the cremasteric reflex); the ilioinguinal (anterior); the posterior scrotal (pudendal).

2 · Peri-operative principles and access

Preparation and dartos closure prevent most complications.

- Shave with a **razor on the day of surgery**. American Urological Association (AUA): a a single IV cephalosporin or clindamycin **only with risk factors**.
- Pre-incisional 0.5% lidocaine (no adrenaline) reduces pain.
- Benign: scrotal incision (avascular median raphe, or transverse). **Possible malignancy**: inguinal incision, clamp the cord before handling the testis, frozen section, and spare it if benign.
- **Always close the dartos**, because most haematomas come from inadequate dartos closure.

3 · Scrotal wall: excision, scrotectomy, reconstruction

Fournier gangrene is the main cause of skin loss.

- **Fournier gangrene** is the commonest indication for partial scrotectomy. Give fluids and broad-spectrum antibiotics, and **debride urgently**, because it spreads **2–3 cm/h**; mortality is about **20%**. Re-debride every **24–48 h**; anorectal extension may need a colostomy.
- Scrotal violation in non-seminomatous germ cell tumour (NSGCT) calls for partial scrotectomy of the scar.

- If more than **50–60%** of skin is lost, do not reconstruct at once; place the testes in thigh pockets, under saline gauze or under vacuum dressing (VAC).

ALGORITHM

1. If <50% and tension-free, then close primarily.
2. If <50% with tension, then local scrotal advancement or secondary intention.
3. If >50%, then VAC, then stretched scrotal advancement or delayed primary closure.
4. If no local tissue, then fasciocutaneous thigh flap; meshed split-thickness skin graft (STSG) last.

KEY POINT

The scrotum works because it is lax. Graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor.

4 · Hydrocele

Choice of repair trades recurrence against haematoma.

- **Congenital** hydrocele comes from a patent processus; about **80%** resolve by 18 months. **Acquired** hydrocele reflects a production–absorption imbalance, in men over 40.
- Treat for discomfort, cosmesis or malignancy; pyocele urgently. **Colour Doppler ultrasound first** excludes tumour and decides scrotal versus inguinal route.

Technique	Steps	Best for
Lord plication	Small opening; radial plicating sutures; no dissection	Thin-walled, smaller; least haematoma, no drain
Jaboulay (Winkelman)	Excise part of parietal layer, evert remainder behind testis	Thin, floppy sacs
Bottleneck	Trim to 2-cm cuff, tack edges	Excisional
Excision	Excise sac, oversew edges	Thick, large, multiloculated; lowest recurrence

KEY POINT

The more tunica excised, the lower the recurrence but the more haematoma; plication trades a little recurrence for a dry field. Recurrence after any successful repair is **0–6%**.

- **Aspiration and sclerotherapy** suit the unfit or surgery-averse; aspiration alone recurs, so add a sclerosant (doxycycline, tetracycline, alcohol, polidocanol). Check the fluid for sperm.
- With **retreatment**, resolution is **85–96%**. Results are poor if multiloculated or more than **750 mL**.

5 · Spermatocele and epididymal surgery

Epididymal surgery can obstruct sperm transport.

- Operate on a spermatocele only for symptoms, and **counsel about obstructive azoospermia** (corpus/cauda). Dissect to the stalk and ligate it; unroof if inseparable.
- Epididymal tumour: suspected malignancy, inguinal route; proven benign, scrotal microsurgical excision.
- **Epididymectomy** gives **poor results for chronic epididymitis** (>6 weeks), so it is a last resort; it works better with a **palpable lesion**. Ligate the efferent tubules at the caput and

the stumps; orchidectomy may follow.

6 · Varicocele repair

Approach decides recurrence, hydrocele and pain.

- P2 Surgical management of bilateral varicocele — *~15% of men; Palomo, laparoscopic, microsurgical subinguinal (lowest recurrence and hydrocele, gold standard) and embolisation compared; microsurgical steps*

Not in the paper answer:

- †Palomo (high) ligation misses cremasteric and external spermatic collaterals, and often takes artery and lymphatics, causing hydrocele.
- †Ivanissevich (inguinal) repair opens the external oblique, so pain is greater. Subinguinal repair has no fascial incision, so pain is least, but it meets more veins and small arteries.
- †Embolisation suits recurrence after surgery; its risks are coil migration and radiation.
- For orchialgia, **microsurgical repair** gives more complete relief.

7 · Vasectomy

Counselling and occlusion method decide success.

- **Counsel**: not sterile at once, so use contraception until post-vasectomy semen analysis (PVSA). Repeat vasectomy is needed in **<1%**; pregnancy occurs in about **1 in 2000** even after an azoospermic PVSA; chronic pain affects about **1–2%**.
- Give **no routine antibiotics**. Topical cream is never the sole anaesthetic.
- **Isolation** determines morbidity; **occlusion** determines efficacy.

Feature	Conventional (CV)	Minimally invasive (MIV)	No-scalpel (NSV)
Skin opening	1.5–3 cm	≤ **10 mm**	Puncture <10 mm
Status	Difficult cases	**Recommended**: less pain, bleeding, infection	Original MIV (Li Shunqiang, China, 1974)

- Open on the **straight vas** at or above the penoscrotal junction, because the convoluted vas is harder now and at reversal.
- NSV (ring clamp, sharp mosquito) defines only the isolation step.

AUA-recommended occlusion (failure ≤1%)	Note
Mucosal cautery (MC) and fascial interposition (FI)	FI = internal spermatic fascia over one end
MC alone (no clips/ligatures)	Cautery of lumen 0.5–1.5 cm
Open-ended: abdominal end MC and FI, testicular end open	Less congestive epididymal pain; granuloma linked to better reversal

KEY POINT

Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division is required; segment excision is optional (**1 cm** is adequate).

ALGORITHM

1. Do PVSA at **8–16 weeks** (after at least 10–20 ejaculations).
2. If azoospermia or rare non-motile sperm (RNMS, ≤ **100,000/mL**), then stop contraception.
3. If motile sperm, then repeat at 6 months; if still motile, then it is a failure, so repeat the vasectomy.

- **Early failure** is technical (one vas missed), seen on PVSA. **Late**: recanalisation after successful PVSA (about 1 in 2000).
- Complications: haematoma, infection, and **sperm granuloma**, which is expected rather than a complication (NSAIDs; excise if it persists, <5%). Post-vasectomy pain affects about **2%**; if a cord block helps, then denervation or reversal.

8 · Orchidectomy and prosthesis

Route differs for cancer, trauma and pain.

- **Bilateral simple orchidectomy** treats hormone-sensitive metastatic prostate cancer. It is cheaper than androgen deprivation therapy (ADT), with less gynaecomastia, but has no proven survival benefit.
- **Unilateral**: trauma or non-salvageable torsion; for orchialgia, only after all conservative measures.
- **Epididymis-sparing** orchidectomy (under the microscope) gives a better contour; †subcapsular orchidectomy has the same aim.
- **For orchialgia, use the inguinal route**, because it removes more of the proximal ilioinguinal, iliohypogastric and genitofemoral nerves. If pain persists, resect the genital branch.
- **Prosthesis**: offer it to every man lacking a testis, before cancer orchidectomy. Pierce the fill port ≤ **5 times**, never after insertion. A **high scrotal subdartos** pouch has least infection and erosion; trans-scrotal placement extrudes more.

9 · Testicular torsion and fixation

Speed saves the testis.

- †**Acute torsion**: if suspicion is high, explore without waiting for imaging. Detorse and assess, then fix or remove; **always fix the contralateral testis**.
- Intermittent torsion (pain <2 h) and postpubertal retractile testis need elective orchidopexy.

Suture fixation	Dartos pouch
Albuginea to dartos (3-0 absorbable). Re-torsion is possible even with non-absorbable sutures. Transparenchymal sutures cause inflammation, abscess and spermatogonia loss (animals)	Subdartos pouch with a purse-string at the cremaster exit; no transparenchymal suture, so **preferred when fertility matters**

10 · Chronic scrotal pain (>3 months)

Find a cause first; nerve blocks guide surgery.

ALGORITHM

1. History, examination, duplex ultrasound; if a cause is found, then treat it.
2. If none, then exclude extrascrotal causes (image abdomen, pelvis, spine, hip); treat prostatitis or pelvic floor causes.
3. If non-specific, then give support, NSAIDs, tricyclics or gabapentin for ≥3 months.
4. If epididymal, then epididymectomy.
5. If testicular or post-vasectomy, then do a cord block: if relief >**50%**, then microsurgical denervation (or reversal); if <50%, then pain clinic, with inguinal orchidectomy last.
6. If post-hernia, then ilioinguinal block; if relief >**90%**, then neurolysis.

- **Microsurgical denervation of the spermatic cord (MDSC)**: through a subinguinal incision, divide the ilioinguinal nerve and genital branch. Use a micro-Doppler to **preserve testicular and cremasteric arteries, lymphatics, and the vas with its vessels**; divide everything else.
- The **trifecta nerve complex** (cremaster, perivasal tissue, posterior periarterial tissue/cord lipoma) is all divided.

11 · Complications of scrotal surgery

Haematoma dominates.

- **Haematoma** is commonest; drain it if large, infected or enlarging. Bleeding after orchidectomy means a slipped cord ligature, so re-ligate subinguinally. A falling haematocrit suggests a **retroperitoneal bleed**, so extend into the retroperitoneum.
- Iatrogenic epididymal or vasal obstruction causes azoospermia.

12 · Seminal vesicles

Rare disease; imaging and embryology explain most.

- The seminal vesicles (SV) are Wolffian derivatives that bud at about **12 weeks**; they make about **60%** of seminal fluid. Relations: bladder in front, rectum behind, **neurovascular bundles lateral to the tips**, ureter anterolateral.
- **MRI is most accurate**. On transrectal ultrasound (TRUS), suspect obstruction if the SV is >**1.5 cm** or the ejaculatory duct >**2.3 mm**; confirm by **aspiration showing sperm**.
- Cysts are rare, congenital or acquired from ejaculatory duct obstruction (EDO). **Zinner syndrome** is an SV cyst with **ipsilateral renal agenesis** and EDO, often with an ectopic ureter into the cyst; †they share a mesonephric origin.
- Observe or aspirate small cysts. **Transurethral unroofing** is preferred for cysts, EDO and stones; large symptomatic cysts need laparoscopic/robotic excision. Stones or haemospermia: seminal vesiculoscopy and holmium; re-obstruction is common.

Benign	Malignant
Cystadenoma, papillary adenoma, hydatid cyst, amyloid	Adenocarcinoma, mixed epithelial–stromal, phyllodes, sarcoma, carcinoid, seminoma, yolk sac

- Image with MRI; if inconclusive, do a TRUS biopsy. †Exclude secondary invasion (prostate, bladder, rectum), which is far commoner. Adenocarcinoma needs radical excision and lymphadenectomy, with or without adjuvant therapy.

13 · Surgical approaches to the seminal vesicles

Choice depends on laterality, size and ureteric anomaly.

Approach	Use	Main risk
Transvesical transtrigonal	Large benign cyst; ectopic ureter into cyst	Urinary leak, ureter
Perivesical	Children, large cyst with or without nephroureterectomy	Ureter, vas
Transperitoneal (laparoscopic/robotic)	Bilateral lesions; dominant, because it opens neither organ	Access injuries
Perineal	At perineal prostatectomy	Neurovascular bundle
Transcoccygeal	Hostile abdomen/perineum	Rectum

- Injuries: rectum, two-layer repair, colostomy if spillage; ureter, reimplant (with or without psoas hitch/Boari) over a stent; neurovascular bundle, stay medial.

★ Putting it together

A · Testicular mass: the first week

The order of steps protects staging and fertility.

ALGORITHM

1. Ultrasound (US).
2. Tumour markers AFP, hCG and LDH **before** orchidectomy; sperm bank.
3. **Radical inguinal orchidectomy** (never trans-scrotal).
4. Repeat markers (AFP half-life 5–7 days, hCG 1–1.5 days); CT abdomen–pelvis and chest (*see Ch 79, 80*).

B · Germ cell tumour by stage

Histology and stage set the treatment.

Stage	Seminoma	Non-seminoma (NSGCT)
Clinical stage (CS) I	Surveillance (occult spread 10–15%); single-dose carboplatin	Low risk: surveillance. High risk (lymphovascular invasion, embryonal-predominant): surveillance, BEP ×1 or nerve-sparing RPLND
II	≤3 cm: radiotherapy, chemo or RPLND; >3 cm: chemo	Normal markers and small nodes: RPLND; otherwise chemo
Advanced (IGCCCG)	Good: BEP ×3 or EP ×4. Intermediate/poor: BEP ×4 (VIP ×4 if lung concern). Full-dose cisplatin	
Residual mass	>3 cm: FDG-PET (≥6 weeks post-chemotherapy)	Any >1 cm: post-chemotherapy RPLND (teratoma)

C · Penile cancer: primary and groin

Treat the primary conservatively; stage the groin by risk.

ALGORITHM

1. **Primary:** if penile intraepithelial neoplasia (PeIN), then topical 5-FU/imiquimod, laser or glans resurfacing. If invasive, then organ-sparing surgery, glansectomy, or partial/total penectomy; margins <1 cm are acceptable (*see Ch 82*).
2. **cN0:** if low risk (pTis/pTa/pT1a), then surveillance; if intermediate/high, then dynamic sentinel node biopsy (DSNB) or modified inguinal dissection (ILND); if positive, then radical.
3. **cN+:** if FNA positive, then bilateral radical ILND; if cN3, then neoadjuvant TIP first. Add pelvic dissection if ≥2 inguinal nodes or extranodal extension (ENE) (*see Ch 84*).

D · Anterior urethral stricture

Match the operation to site and length.

Stricture	Treatment
Bulbar <1.5 cm, first time	Dilation or DVIU once; if it recurs, urethroplasty
Bulbar 1–2 cm (3–5 cm proximally)	Excision and primary anastomosis (EPA)

Stricture	Treatment
Longer bulbar	Buccal mucosa graft (BMG) onlay, with or without augmented anastomosis
Penile	Onlay graft/flap; staged if hostile bed, lichen sclerosus, failed hypospadias
Panurethral, frail	Perineal urethrostomy
Pelvic fracture urethral injury	Suprapubic catheter, then delayed progressive perineal anastomosis

In retention, place a suprapubic catheter; never blind-dilate (*see Ch 86*).